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Studying the impact of Constrained Primordial Non-Gaussian Fields in the Cosmic Web

MPhys Project Report

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Abstract

Identifying and studying the effects of early universe theories on cosmological structure formation can serve as a useful probe of these models. The Caustic Skeleton method, together with the Hoffman-Ribak method for constrained Gaussian random fields (GRFs), has the potential to do just this, providing a method to study the evolution of specific structures in the cosmic web. However, this methodology has been applied only to Gaussian fields, leaving many potential early-universe signatures uncovered. This project aims to extend this methodology to the study of non-Gaussian fields in 2 and 3 dimensions. It presents, for the first time, a method to generate constrained Local non-Gaussian fields, presenting the constraining conditions for a 2D filament and a 3D cosmic wall. Secondly, it shows how a generic non-Gaussian field can be described in terms of a combination of linear and quadratic constraints on the GRF used to build it. Lastly, it develops two approximations that make these constraints compatible with the Hoffman-Ribak algorithm. This opens the door to future studies that can quantify statistical effects of non-Gaussianities in large-scale structure formation.

Supervisor: Dr. Job Feldbrugge

Personal statement

In this project, I aimed to study the theoretical aspects of non-Gaussian fields and the current constrained methods used for Gaussian Random Fields (GRFs) to develop new methods for constraining non-Gaussian fields. This was mostly a mathematical study, so that future projects can use what I described to investigate the properties of different structures in the cosmic web under the influence of different types of primordial non-Gaussian fields. This forced me to develop a completely new understanding of the Caustic Skeleton method, to learn how to apply constrained methods to GRFs and all the associated machinery needed to sample these realisations, as well as to develop a deep understanding of how to describe a generic non-Gaussian field, in order to develop the methods I was looking for. While I hope my report speaks for itself, here I will outline how my project developed throughout the year, the extra tasks I needed to complete that didn't make it into the report, as well as the wrong turns I took along the way.

During my first month, between mid-September and mid-October, I divided my time between learning constrained Gaussian random field theory and developing my own Hamiltonian Monte Carlo sampler algorithm, since my supervisor knew I would need to perform sampling for this project and wanted me to be able to understand how this process worked so I could use an optimised algorithm if it came to it. Developing this method from scratch not only helped me better understand how samplers work, but also influenced how I wrote my constraints, since I would look for a way to try to reduce the number of variables the sampler would have to work with.

In my second month, I focused on implementing the mathematical constraints on the known results for a constrained GRF before attempting the same for the Local type of non-Gaussian Fields. This involved me learning how to work with Mathematica for the first time, learning how the field invariants method works and then trying to write the equations for it in Mathematica. I started by doing this for the 2D case. Once I managed to reproduce all the known results for the Gaussian fields, I proceeded to implement the results for the local type non-Gaussianities. Then, until the end of November, I spent the rest of my time testing whether this result was correct, using limiting cases and expansion methods, before moving on to building a Julia simulation of the constrained field. In the meantime, I started researching the formalism for generic non-Gaussianities and trying to find papers that would make it easier to apply the methods I developed for the local type to a more general non-Gaussian field.

Over December, I took a break to prepare for my exams. Once this was done, I spent the winter break working on building working simulations of a 2D constrained Gaussian Random Field. For this, I started by using the details described in my supervisor's paper, developing my own implementation before comparing it with examples my supervisor later provided. This also involved rederiving all the correlation and rotation matrices presented in previous papers to ensure I had the correct values. Unfortunately, my implementation had some unexpected bugs and fixing them took longer than I expected. It wasn't until mid-January, after I met back with my supervisor, that I managed to get a working constrained GRF 2D

simulation. Simultaneously, I used the last week of the winter holidays to start writing the introduction for my report.

The rest of January was spent working on generalising the simulation I had for a GRF to the local types of non-Gaussianities, whose conditions I had derived earlier. This led me to find some more bugs and to further refine my code, and I was able to generate the final results by the first week of February.

Once I had the results confirming this methodology, I generated the conditions for the Local non-Gaussianities for 3D as well. These were my first two novel results. I had managed to generate, for the first time, the constraint conditions for a Local type non-Gaussian field. Once this was done, I focused on doing the same for the general non-Local non-Gaussianities.

For the rest of February and all the way towards mid-March, I attacked this problem. Unfortunately, as with many research projects, I took several wrong turns before finding the right approach. Most of these wrong approaches were aimed at finding a way to linearise the quadratic dependence of the non-Gaussian field in order to be able to use it with the Hoffman-Ribak algorithm. These involved trying to treat the Fourier operators as real-space operators, exploring whether there was a way to apply the Hoffman-Ribak method for non-Gaussian fields, or if turning spatial derivatives into functional derivatives would help linearise the field. During this time, I started splitting my time between report writing and putting my methodology and results on paper.

While all these methods eventually failed, they helped me reach my final two conclusions: that we can only get exact constraints for the non-Gaussian field in terms of a quadratic operator on the initial GRF, and thus the Hoffman-Ribak method can only be applied if we approximate the non-Gaussian field in terms of linear factors. In the end, I found two alternative methods to approximate the field for this purpose. However, due to the wrong turns I took during this exploration, I didn't have the time to implement a full example with a known non-Local non-Gaussian Random Field, as I would have wished to do. However, I did manage to fulfil my goal of developing a method for producing constrained realisations of a generic non-Gaussian random field, having outlined the steps required to achieve this.

Overall, this project has allowed me to become a better physicist and has helped me further develop my analytical skills to find new solutions to problems that haven't been attempted before, and, above all, has taught me to persist even when these fail until I eventually find something that works.

Acknowledgments

I would like to thank my supervisor, Job Feldbrugge, for all the help, guidance and patience during this project. His advice and explanations were invaluable not only in getting this project off the ground but also in helping me improve as a physicist. Lastly, I would like to thank my friends and family for their continuous support.

The idea for this project originated from the results of my Senior Honours Project, during which I showed how to study specific unconstrained realisations of local, orthogonal and equilateral non-Gaussianities in the 2D cosmic web. This project took this a step further, delving deeper into the Caustic Skeleton method to develop methods to generate constrained realisations of generic non-Gaussian fields in both 2 and 3 dimensions. So while my previous project gave me some familiarity with the Caustic Skeleton method, I had to learn how this can be used in 3D, as well as how to identify additional conditions for the centre of a filament. All the aspects of constrained Gaussian Random Fields were new to me, as well as the formalism for a generic non-Gaussian field. The results presented in this report present the next step in the quest to study primordial non-Gaussianities using the caustic skeleton method, providing a methodology for examining the effects of different types of non-Gaussianities in specific structures of the cosmic web.

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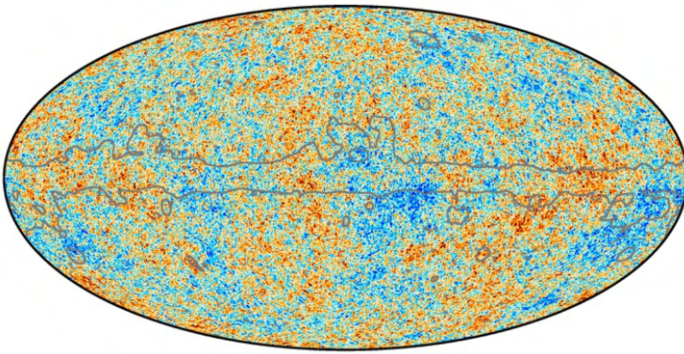
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1 Introduction

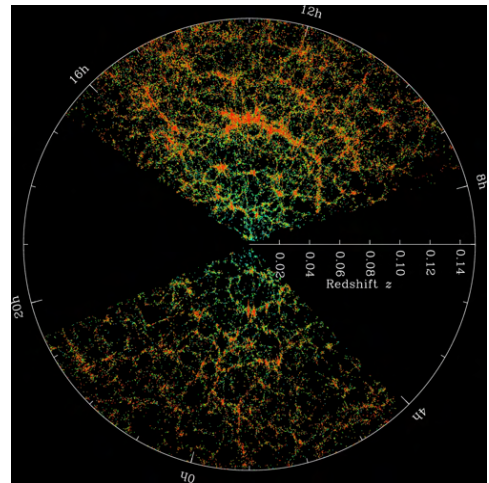
The study of cosmological structure formation seeks to answer some of the most basic questions we have about our universe, helping us understand how the universe evolved over time to create the structures we observe today. One of the key questions we ask is whether we can use the universe as we see it today to learn more about the early universe and how it has influenced the later formation of structures. In this project, we aim to develop the methods that would allow us to answer this question.

However, we must first develop a broader understanding of the universe, its evolution and how we can study it. If we want to study the early universe and its composition, we must start by studying the Cosmic Microwave Background (CMB). The CMB is the radiation emitted when the early universe became transparent to light, carrying a "picture" of the universe at that time, the earliest information we can get about the universe. Figure 1a shows us a temperature map of the CMB as measured by the Planck collaboration. This picture shows how the photons from this primordial time have cooled off to very low temperatures, but still present small fluctuations resulting from the structures that existed at that time.

Studying the properties of this isotropic background is a powerful tool to understand the physics of the early universe. While initial surveys only sampled this background's overall temperature, at around 3.5 K [1], further surveys have all observed a temperature of about 2.725 K [2], and small anisotropies of the order of 10^{-5} K. This substructure was first found with the Cosmic Background Explorer (COBE) survey [3], which identified the CMB dipole and was further studied in the Wilkinson Microwave Anisotropy Probe (WMAP) [4] and the Planck collaboration [2, 5], which produced the most precise mapping of the CMB so far.



(a) The CMB as measured by Planck [2].



(b) The Universe as seen by the SDSS collaboration, each dot in this picture is a galaxy [6].

Figure 1. The CMB and the Large Scale Structure, two pieces of the same puzzle to understand primordial non-Gaussianities

Current measurements of the CMB seem to indicate this is very close to a Gaussian Random Field, a multidimensional generalisation of a Gaussian distribution (a field where every single point is the result of a Gaussian distribution draw). However, recent studies still leave a margin for the existence of small amounts of non-Gaussianity in this field [7].

Studying the nature of these primordial non-Gaussianities and finding observational constraints on them may prove essential to understanding early universe physics, since different theories will predict different types of non-Gaussianities. For example, different inflation theories produce different non-Gaussian signatures [8, 9]. Distinctive features of the primordial density perturbations are also predicted by cyclic models [10] or the CPT-symmetric theory [11, 12]. Identifying the nature of the initial perturbations we observe in the early universe may then prove a key observable for distinguishing which theories may correspond to our reality.

While studying these non-Gaussianities directly from the CMB is a valid and valuable approach, this avenue has already been explored with high precision. While this provides us with invaluable information, the CMB is a 2D field, carrying less information than, for example, a 3D field, and there are other non-Gaussian contaminants that cannot be eliminated from its analysis [13].

An alternative method to study primordial non-Gaussianities is to analyse the effects that these might have on structures in the later universe. The primordial density fluctuations, as shown in the CMB, evolve through gravity in a non-linear way to form what we call the Cosmic Web, the large-scale structure of our universe we observe nowadays, formed by overdense regions such as walls, filaments and clusters, and voids, regions of underdensity. We can map this large-scale structure by studying the distribution of galaxies at the largest scales possible, as shown in Figure 1b, where the web-like structure of overdense regions separated by voids is clearly visible. Current large-scale structure surveys have provided measurements of the distribution of galaxies with unprecedented precision. The main developments in this field have come from the two-Degree-Field Galaxy Redshift (2dF) [14], the Sloan Digital Sky Survey (SDSS)[15], the Dark Energy Survey [16] or the Dark Energy Spectroscopic Instrument (DESI) [17] surveys and we will be seeing even better measurements coming from future missions like the Euclid collaboration [18], giving us comprehensive surveys of galaxy distribution at large scales.

Studies of the CMB and large-scale structure formation are thus two pieces of the same puzzle to understand our universe. The primordial anisotropies in the CMB can not only be mapped to different early universe theories, but are also directly related to the large-scale structure we observe nowadays. It is thus possible to learn more about the early universe by studying the nature of these initial perturbations through the analysis of the Cosmic Web.

In the last few years, several simulations capable of modelling the evolution of the universe from primordial perturbations have been developed. Several simulations have been successful in using only Gaussian primordial perturbations to model this evolution, an example of these are the AbacusSummit cosmological N-body simulation suite [19] and the IllustrisTNG or MillenniumTNG hydrodynamical simulations [20, 21]. Similarly, several

simulation collaborations have now managed to successfully produce large-scale structure simulations implementing different types of primordial non-Gaussian density perturbations, such as the Quijote-PNG [22], the GENGARS [23], and the PNG-UNITsims [24] simulations. However, while current simulations can help us understand how the early universe’s quasi-Gaussian conditions form the current cosmic web, most surveys don’t provide us with an easy way to study specific structures of the cosmic web, for example, filaments, which makes it difficult to study in detail the impacts of different types of non-Gaussianities.

The caustic skeleton method [25, 26] provides a more systematic way to study the Cosmic Web. Through this method, we can identify the different structures of the cosmic web by studying the properties of the primordial gravitational field. Using this analytical method, we can then consider which properties a primordial field should obey for us to observe a given structure, and derive analytical expressions for this. Using these conditions, we can impose them on the initial field we use to simulate the primordial density fluctuations, generating constrained fields.

While this technique has been applied to the study of Gaussian initial conditions [27, 28], there hasn’t been any generalisation of this procedure for non-Gaussian random fields. In this project, we aim to develop a novel method for studying different types of non-Gaussian Fields using the apparatus developed for the Caustic Skeleton method to identify distinct structures. In this report, we have derived a method to generate constraints for a Local type non-Gaussian field, producing, for the first time, constrained conditions for a 2D filament and a 3D cosmic wall. Secondly, we show how the constraints for a generic non-Local non-Gaussian field can be determined in terms of both linear and quadratic expressions of the initial Gaussian Random field, which are incompatible with the constrained method we use. Lastly, we show how these constraints can be approximated in order to make it compatible with the rest of our methodology, presenting two approaches that can be used to achieve this.

In order to arrive at these conclusions, we will start by first discussing the Caustic Skeleton method and how it can be used to identify and study different structures in the cosmic web in section 2. Secondly, in section 3 we will introduce the formal definition of Gaussian Random Fields, their properties and how to generate their constrained realisations. In section 4, we show how these methods can be used to generate the constraint conditions for a Gaussian example, before showing how this method can be generalised to non-Gaussian fields in section 5. Section 5 will be divided into 3 subsections. First, we will discuss a method for generating constrained realisations of Local non-Gaussian fields, the simplest case of a non-Gaussian Field realisation, providing the caustic conditions for both 2D and 3D. Lastly, in sections 5.2 and 5.3, we discuss how we can define a generic non-Gaussian field and why its quadratic nature in the field will force us to consider only approximations for the constrained conditions of these fields, giving two methods that can be explored to achieve this. In section 6, we provide an overview of our results and outline next steps for future researchers to continue this project.

2 The Caustic Skeleton

The main goal of this project is to present a method to study the effects of non-Gaussianities in specific structures of the cosmic web. In our specific case, we will be focusing on 2D filaments and 3D cosmic walls. But how can we geometrically define these elements in the first place? Similarly, how can we simulate the evolution of structure to establish a connection between the early-universe conditions, which we simulate as our initial conditions, and the late-universe structures we want to study?

In this section, we aim to present the main concepts that underpin the Caustic Skeleton method and allow us not only to establish a connection between primordial universe conditions and the structures in the late universe, but also to identify the properties an early universe simulation must obey for specific structures to form.

2.1 Structures in the Cosmic Web

As we discussed before, on large scales, our universe shows a web-like structure of dense regions separated by empty voids. To understand how these structures evolve, we need to identify and classify them. To do this, we must first understand how different structures form and evolve.

At large scales, we can understand the evolution of the matter of the universe as the evolution of a dark matter fluid under gravity [25]. Given this, we can use a Lagrangian fluid-dynamics approach to study the evolution of the universe. By this, we mean that we consider each point in space as a mass element moving with the fluid.

At large scales, we can consider our universe as obeying the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, $g_{\mu\nu} = \text{Diag}(-1, a^2(t), a^2(t), a^2(t))$, where $a(t)$ is the scale factor. Similarly, for this project, we assume the standard model of cosmology, as defined by the Friedmann equation for a Λ CDM universe:

$$H(t) = H_0 \sqrt{\Omega_\Lambda + \Omega_m a^{-3} + (1 - (\Omega_\Lambda + \Omega_m)) a^{-2}}, \quad (1)$$

where $H(t) = \dot{a}/a$ is the Hubble parameter, and $\Omega_\Lambda = \rho_\Lambda/\rho_c$ and $\Omega_M = \rho_m/\rho_c$ are the dark energy and matter density parameters, with the critical density defined as $\rho_c = \frac{3H_0^2}{8\pi G}$.

On large scales, we can treat gravity in the Newtonian limit, assuming the dark matter fluid obeys Newton's gravitational equation in an expanding universe. This means we can describe the evolution of this fluid using Poisson's equation and Newton's law [29]:

$$\nabla_x^2 \Phi = 4\pi\bar{\rho}\delta(t), \quad (2)$$

$$\ddot{\mathbf{x}} + H(t)\dot{\mathbf{x}} = -\frac{1}{a(t)^2}\nabla_x\Phi. \quad (3)$$

These equations are non-linear and cannot be solved exactly. Instead, these can be solved iteratively using N-Body simulations to study the non-linear evolution of the cosmic web, as seen in [30] and [31]. Our goal is to understand how to analytically identify different regions of the cosmic web simulated by this type of simulations.

In Lagrangian fluid dynamics, we consider the evolution of the universe as moving from an initial (Lagrangian) space, defined by positions $\mathbf{q}$ to a final (Eulerian) space, defined by coordinates $\mathbf{x}(t)$. To define this evolution, we can use the Zel'dovich approximation [32], which assumes that we can approximate the initial states of structure formation as behaving linearly, thus providing a linear map between the two spaces:

$$\mathbf{x}_t = \mathbf{q} + \mathbf{s}_t(\mathbf{q}) = \mathbf{q} - D_+(t)\mathbf{s}(\mathbf{q}), \quad (4)$$

where $\mathbf{s}_t$ is the displacement between point $\mathbf{q}$ and point $\mathbf{x}$ at time t and can be defined as $\mathbf{s}_t(\mathbf{q}) = -D_+(t)\mathbf{s}(\mathbf{q})$, where $\mathbf{s}(\mathbf{q})$ is the time independent displacement field and the $D_+(t)$ is the growth factor that describes the time evolution at time t . This equation describes how a mass element situated at $\mathbf{q}$ moves into a final position $\mathbf{x}(t)$. While this approximation is linear, it captures the non-linear evolution of the cosmic web for small growth factors (up to $D_+(t) \approx 1$) [33].

From this approximation, we can now try to understand how the regions of overdensity in the universe came to be. This happens because, while the initial gravitational field of the universe is very homogeneous, it presents small fluctuations (as we see in the CMB in Figure 1a). These fluctuations lead the mass elements at each point to move towards places with a slight overdensity. Over time, this evolution will compound, and in our mapping from Lagrangian to Eulerian space, we will see that mass elements that corresponded to different coordinates in Lagrangian space will correspond to the same point in Eulerian space. We use this correspondence between Eulerian and Lagrangian spaces to identify different regions in the cosmic web, we call these stream regions. For example, a single stream region corresponds to a 1:1 correspondence from Lagrangian to Eulerian spaces, and a 3-stream region corresponds to a 1:3 correspondence. Identifying different regions in the cosmic web has now become a matter of mapping the boundaries of multistream regions.

A schematic example of this is shown in Figure 2, which depicts the different regions of a 1D dark matter sheet. As we can see, as the sheet deforms (moves into Eulerian space), we start to have multiple points that correspond to the same position in Eulerian space. Another way to see this is by deforming a sheet of paper. When the paper is unfolded, each point on the sheet corresponds to a separate coordinate, however, as we fold it, different points on it roughly correspond to the same spatial point.

To understand how this mapping can be done, we must introduce a formalism for treating our fluid. First, going back to our definition of the linear mapping in equation 4, we defined our displacement potential as two linearly independent functions. Now it's important to further define what they mean. The growth factor can be fully determined by Einstein's equations and thus depends only on cosmological parameters, whereas the displacement field is a vector potential, and can be defined as [26]:

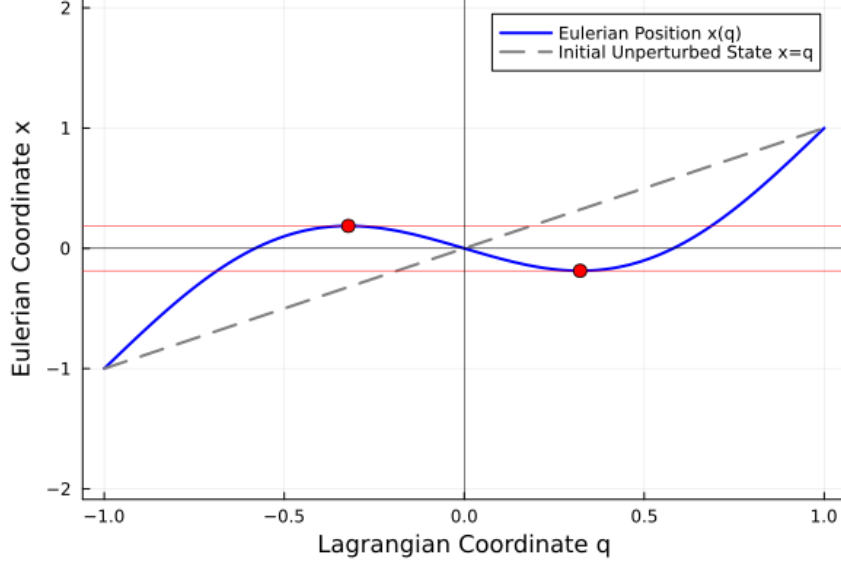


Figure 2. 1D representation of a Dark Matter sheet undergoing evolution under the linear mapping described in equation 4. The blue line represents a sheet with a 1-stream and a 3-stream region, where the red lines represent the boundaries of the 3-stream region. The red dots represent the boundary of the 3-stream region in the dark matter sheet, these are the points of infinite density we try to identify when looking for the boundaries of different regions.

$$\mathbf{s}(\mathbf{q}) = \nabla_{\mathbf{q}}\psi(\mathbf{q}) , \quad (5)$$

where the $\psi(\mathbf{q})$ is the displacement field, which can be determined in terms of the initial gravitational potential ϕ_0 , and the current energy density of the universe, Ω_0 :

$$\psi(\mathbf{q}) = \frac{2}{3\Omega_0 H_0^2} \phi_0(\mathbf{q}) . \quad (6)$$

To study different stream regions, we are interested in studying the evolution of the density field in Eulerian space (higher stream regions will have more mass elements, thus be denser). We can obtain this from mass conservation, which implies that $\rho(\mathbf{x}, t)d\mathbf{x} = \bar{\rho}(t)d\mathbf{q}$, where $\bar{\rho}(\mathbf{q})$ is the average density of the early universe. From this, we can see that the density field in Eulerian space can be described as:

$$\rho(\mathbf{x}, t) = \bar{\rho}(\mathbf{q}) \left| \frac{\partial \mathbf{x}}{\partial \mathbf{q}} \right|^{-1} (t) , \quad (7)$$

where $\left| \frac{\partial \mathbf{x}}{\partial \mathbf{q}} \right|$ can be seen as the Jacobian of the mapping between these two fields. We can now introduce one more quantity to further simplify our expression, the deformation tensor of the field (M), described as the gradient of the displacement field with respect to the Lagrangian

coordinates ($M = \frac{\partial \mathbf{s}(\mathbf{q})}{\partial \mathbf{q}}$). Using these definitions, we can now describe the Eulerian density field in terms of the eigenvalues of the deformation tensor at a given Lagrangian position:

$$\rho(\mathbf{x}, t) = \sum_{\mathbf{q} \in \mathbf{x}_t^{-1}(\mathbf{x})} \frac{\bar{\rho}(\mathbf{q})}{\prod_{i=1}^N |(1 + \mu_i(t, \mathbf{q}))|}, \quad (8)$$

where $\mu_i(t, \mathbf{q})$ are the eigenvalues of the deformation tensor and N is the number of dimensions we are dealing with. In this project, we will always start with 2D examples before attempting to generalise our conclusions to a 3D field.

2.2 Shell-Crossing

Shell-crossing is defined as the process of moving between different stream regions. We can use this process to identify the boundaries between these regions. Firstly, we can consider this from the perspective of the density field. The boundaries between different stream regions are by definition infinitesimally small, so when a mass point crosses it, we will have a non-zero mass passing through an infinitesimally small amount of space, which will create a singularity in the density field. Then, using equation 8, we can define what we call the first shell-crossing condition:

$$1 + \mu_{it}(\mathbf{q}_s) = 0. \quad (9)$$

However, while this provides us with a method to identify boundaries, we still don't have a method to identify which structures these are bounding. In other words, we have developed a method of identifying different structures but not of classifying them.

To classify different crossing points, we need to consider again what happens to our mapping when we cross different regions, this time using position mappings rather than the density map. Shell crossing happens at the boundary where 2 points in Lagrangian space start to correspond to the same point in Eulerian space. This point can be described as:

$$\frac{\Delta \mathbf{x}}{|\Delta \mathbf{q}|} = \frac{|\mathbf{x}_t(\mathbf{q}') - \mathbf{x}_t(\mathbf{q}'')|}{|\mathbf{q}' - \mathbf{q}''|} \rightarrow 0 \quad \text{as } \mathbf{q}', \mathbf{q}'' \rightarrow \mathbf{q}_s, \quad (10)$$

where $\mathbf{q}_s$ is called the critical point, the point where the singularity happens in the density field, Figure 3 illustrates how this happens for a 2D sheet. This means that the mapping $\mathbf{x}_t(\mathbf{q})$ at this point becomes discontinuous. In general, we can see this as the partial derivative of $\mathbf{x}_t(\mathbf{q})$ vanishing along the tangential direction of the crossing ($\mathbf{T}$):

$$\left| \frac{\partial \mathbf{x}_t}{\partial \mathbf{q}} \cdot \mathbf{T} \right| = 0 \Leftrightarrow \left| \mathbf{T} + \frac{\partial \mathbf{s}_t}{\partial \mathbf{q}} \cdot \mathbf{T} \right| = 0 \Leftrightarrow |\mathbf{T} + \mathbf{M} \cdot \mathbf{T}| = 0. \quad (11)$$

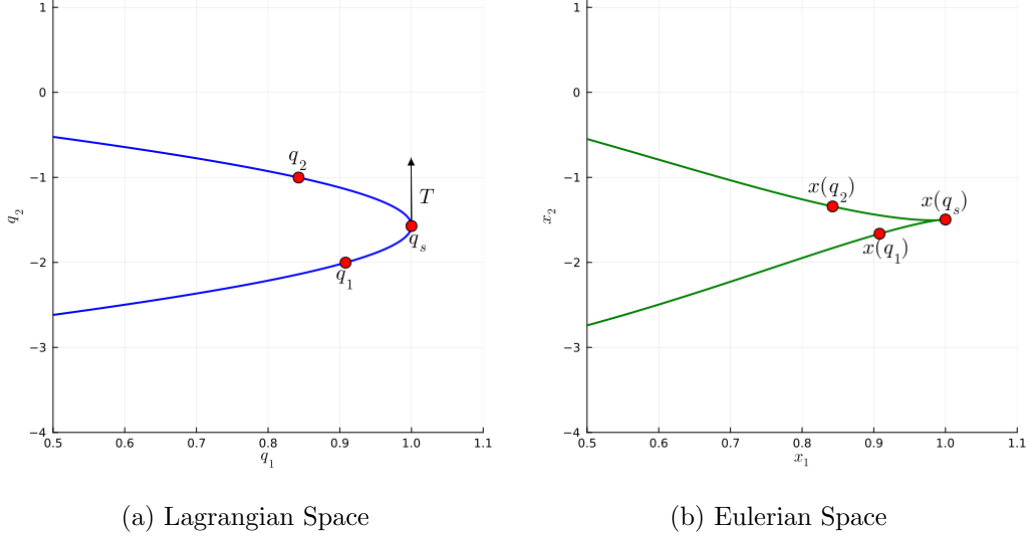


Figure 3. Shell crossing in 2D space. Here we can see how as a 2D sheet in Lagrangian Space, Figure 3a, starts moving into Eulerian space, Figure 3b, and its points start overlapping, it will create a discontinuous region in the point where this region starts happening, here seen at the point $x(q_s)$.

We can simplify this expression by diagonalising the deformation tensor:

$$T + M \cdot \mathbf{T} = (I + M)VV^{-1} \cdot \mathbf{T} = (I + M_d)V^{-1} \cdot \mathbf{T}, \quad (12)$$

where V is the eigenvector matrix and M_d is the diagonalised deformation tensor. Lastly, we can define the tangent vector as the gradient of the eigenvalue field, since this will always point towards the direction of steepest increase, $\mathbf{T} = \nabla \mu_i(t, \mathbf{q})$, giving us a way to write the previous condition as:

$$(1 + \mu_{it}(\mathbf{q}_s)) \mathbf{v}_{it}^*(\mathbf{q}_s) \cdot \nabla v_{it}(\mathbf{q}_s) = 0. \quad (13)$$

From this, we can infer 2 conditions for the occurrence of shell-crossing, the one we had already derived from the density field and a second one:

$$1 + \mu_{it}(\mathbf{q}_s) = 0 \quad OR \quad \mathbf{v}_{it}^*(\mathbf{q}_s) \cdot \nabla \mu_{it}(\mathbf{q}_s) = 0 \quad (14)$$

2.3 Caustic Conditions

The use of shell-crossing conditions provides a new tool for identifying the different structures of the cosmic web by studying the behaviour of both the eigenvalue and eigenvector fields. We define caustics as the points where a mass element undergoes shell crossing, these will

Name	Symbol	Caustic Conditions (Additive within same family)	2D Element	3D element
Trivial	A ₁	$1 + \mu_{it}(\mathbf{q}) \neq 0$	single stream	single stream
Fold	A ₂	$1 + \mu_{it}(\mathbf{q}) = 0$	multistream	multistream
Cusp	A ₃	$\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla \mu_{it}(\mathbf{q}) = 0$	filament	wall
Swallowtail	A ₄	$\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla(\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla \mu_{it}(\mathbf{q})) = 0$	cluster	filament
Butterfly	A ₅	$\mathbf{v}_{it}(\mathbf{q}) \cdot (\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla(\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla \mu_{it}(\mathbf{q}))) = 0$	-	cluster
Umbilic	D ₄	$1 + \mu_{it}(\mathbf{q}) = 1 + \mu_{jt}(\mathbf{q}) = 0$	cluster	filament
Parabolic Umbilic	D ₅ ¹	$\det \mathcal{H}_{\mathbf{v}_i \mathbf{v}_j}(\det \nabla \mathbf{x}_t) = 0$	-	cluster

Table 1. The classification of the different caustics. This table presents the conditions that a given point in Lagrangian space must satisfy to correspond to each caustic, and the structures to which these points correspond in both the 2D and 3D cosmic web. The Caustic Skeleton method takes advantage of this correspondence to identify these structures directly from Lagrangian space, providing us with an analytical method to study these in the later Eulerian space[25, 29]. Here $\mathcal{H}_{\mathbf{v}_1 \mathbf{v}_2}$ represents the Hessian in the $\mathbf{v}_i - \mathbf{v}_j$ plane, where $i \neq j$ for all the expressions shown.

correspond to the boundaries between stream regions we have been looking for. We can now use the machinery we set up to identify different caustic conditions, allowing us to simultaneously identify and classify regions corresponding to different structures.

The shell-crossing conditions provide a tool for studying hierarchical structure formation. We start by finding the points in the field that obey the first shell crossing conditions $1 + \mu_{it}(\mathbf{q}_s) = 0$. This can occur in two cases: either only one eigenvalue satisfies this condition, or 2 eigenvalues satisfy it (in 3D, we can have a third case where the three eigenvalues satisfy this, however, we can show that such a case is degenerate with other structures [25]). We can use these two cases to define two caustic families. The A family refers to situations where only one eigenvalue is zero, the D family to cases with 2 eigenvalues. The caustic families work as a basic manifold, under which there can be points that obey further caustic conditions.

We can exemplify this for the 2D cosmic web by considering the A family of caustics. The A₂ caustic corresponds to the boundary of a multistream region, and is the simplest case we can ever have. This can be achieved by imposing $1 + \mu_{it}(\mathbf{q}) = 0$. To obtain the A₃ family, we just need to impose the second shell crossing condition $\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla \mu_{it}(\mathbf{q}) = 0$. The new points that obey both of these simultaneously will then correspond to a filament in the 2D cosmic web. To reach the A₄ caustic, which represents a cluster of matter, we now need an additional condition. This can be seen as finding all the points that obey the A₃ conditions, and whose orientations are perpendicular to the eigenvector field. This new condition can be described as $\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla(\mathbf{v}_{it}(\mathbf{q}) \cdot \nabla \mu_{it}(\mathbf{q})) = 0$. These are all the A-family caustics in two

¹The following D₅ caustic condition is a correction on the condition presented in previous papers and was obtained through private communication with Job Feldbrugge

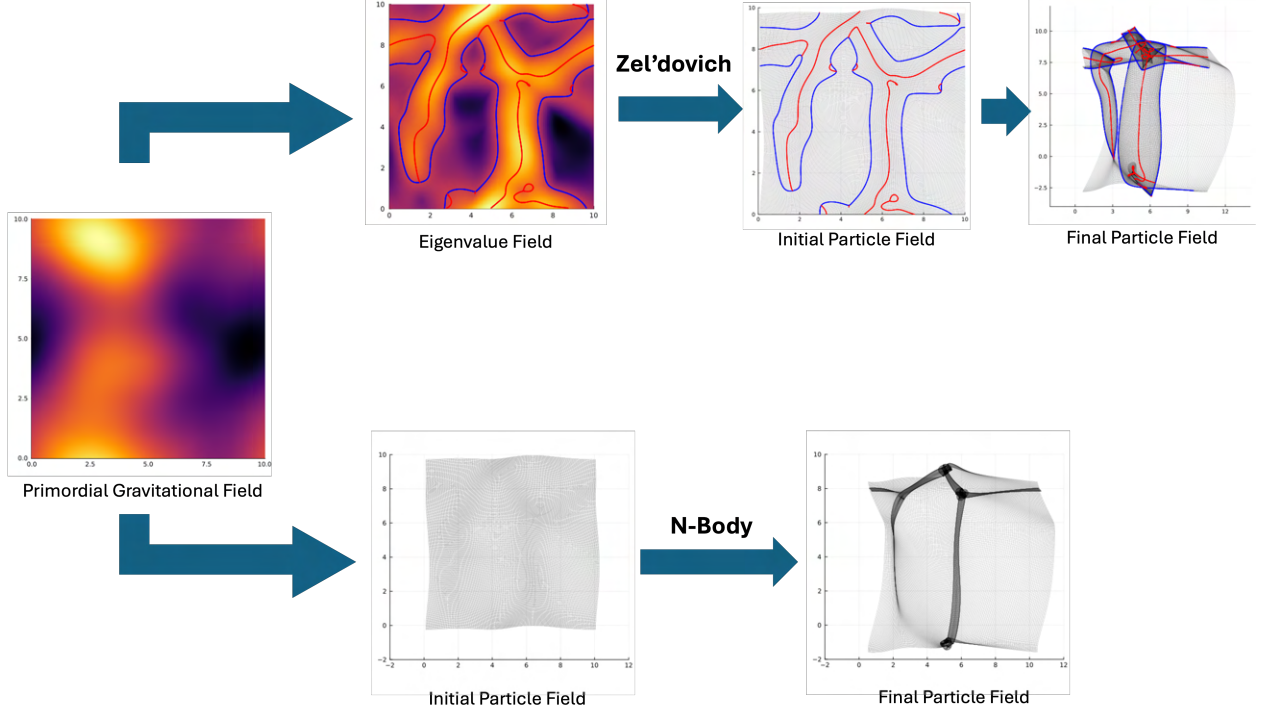


Figure 4. The caustic skeleton method. Starting from a primordial gravitational field, we can use the conditions described above to find the caustics that will become the structures in the cosmic web. Using solely the Zel’dovich approximation to perform the transformation from Lagrangian to Eulerian space, these caustics accurately map the structures in the cosmic web we can observe, and that can be observed in a more accurate N-Body simulation. In the caustic plots, the blue lines represent A_2 caustics, the limits of multistream regions, while red lines represent A_3 caustics, the filaments.

dimensions, but the same logic can be applied to the D caustics and to three dimensions². The full list of caustic conditions for both 2 and 3 dimensions can be found in Table 1.

It is these caustic conditions that underpin the Caustic Skeleton method, enabling us to identify different structures directly in Lagrangian space and then study their evolution into Eulerian space. A schematic of how this process works can be seen in Figure 4. This method thus provides a powerful tool to study the cosmic web directly from the primordial gravitational potential.

In this project, we are interested in the conditions for a 2D filament or a 3D cosmic wall, so that we can force our simulated primordial field to obey them. These are defined as:

$$A_3^i(t) = \{q \in L : 1 + \mu_{it}(\mathbf{q}) = 0 \wedge \mathbf{v}_{it}(\mathbf{q}) \cdot \nabla \mu_{it}(\mathbf{q}) = 0\}, \quad (15)$$

²The nature and classification of the different types of caustics can be further understood using catastrophe theory. However, this is not directly relevant to the objective of this project, and the previous argument provides sufficient intuition for understanding why this method works. A sketch of why this connection is possible and how it allows us to identify the different types of caustics can be found in Appendix A.

where L represents the Lagrangian space.

Next, we need to understand how we can manipulate the primordial gravitational potential in order to force the formation of specific structures, and thus allow us to study specific features of the cosmic web in detail.

3 Random Field Theory

The previous section described a method to study the evolution of the cosmic web based on its initial conditions. We now want to study how to generate this primordial gravitational field in the first place and how we can manipulate its properties. For this, we need to be able to generate both Gaussian Random Fields and constrain them so we can visualise specific structures of the cosmic web and study them in detail.

In this section, we will start by discussing the different properties of Gaussian random fields and how these can be easily generated using computational methods, before discussing how we can take advantage of these same properties to generate constrained realisations of these fields.

3.1 Gaussian Random Fields

A Gaussian Random Field (GRF) is a multidimensional generalisation of a Gaussian distribution to the function space. As such, it can be defined in terms of its probability density functional, the probability of finding a given realisation of a GRF:

$$p[f] = \exp \left(-\frac{1}{2} \int \int d\mathbf{q}_1 d\mathbf{q}_2 [f(\mathbf{q}_1) - \bar{f}(\mathbf{q}_1)] K[\mathbf{q}_1, \mathbf{q}_2] [f(\mathbf{q}_2) - \bar{f}(\mathbf{q}_2)] \right), \quad (16)$$

where $\bar{f}(\mathbf{q})$ is the mean field and K is the kernel. This can be seen as a classical generalisation of a path integral determined by the probability of finding a realisation of a field f :

$$P[f \in S] = N \int \mathbf{1}_S(f) p[f] Df. \quad (17)$$

Thus, we can easily see that Wick's theorem must apply, meaning that all odd moments of the distribution must vanish and that all even ones can be expressed as multiples of the 2-point correlation function, this means all the information about this field is contained in it:

$$\begin{aligned} \xi(\mathbf{q}_1, \mathbf{q}_2) &= \langle (f(\mathbf{q}_1) - \bar{f}(\mathbf{q}_1))(f(\mathbf{q}_2) - \bar{f}(\mathbf{q}_2)) \rangle \\ &= N \int (f(\mathbf{q}_1) - \bar{f}(\mathbf{q}_1))(f(\mathbf{q}_2) - \bar{f}(\mathbf{q}_2)) e^{-S[f]} Df. \end{aligned} \quad (18)$$

Furthermore, knowing we are studying a homogeneous and isotropic universe, we can use the cosmological principle to further simplify this expression, by setting $\xi(\mathbf{q}_1, \mathbf{q}_2) \rightarrow \xi(|\mathbf{q}_1 - \mathbf{q}_2|)$. Adding to that, since we are free to translate the field around, we can also set $\bar{f}(\mathbf{q}) \rightarrow 0$.

Similarly, since, as we discussed before, a GRF can be fully described by its two-point correlation function, it is then important to establish how it relates to the probability density. This can be done by showing that the two-point correlation function is the inverse of the kernel:

$$\int K(\mathbf{q}_1, \mathbf{q}) \xi(\mathbf{q}, \mathbf{q}_2) d\mathbf{q} = \delta^d(\mathbf{q}_1 - \mathbf{q}_2), \quad (19)$$

where d refers to the number of dimensions of the field.

However, many of the properties of a GRF are better understood in the Fourier space, which also helps us develop an effective algorithm to generate this type of field. The most important of these properties is the Power Spectrum, the Fourier transform of the 2-point correlation function:

$$P(\mathbf{k}) = \int \xi(\mathbf{q}) e^{i\mathbf{k} \cdot \mathbf{q}} \frac{d\mathbf{q}}{(2\pi)^d}. \quad (20)$$

Similarly, the relation between the Fourier transform of the Kernel ($\tilde{K}$) and the Power Spectrum is given by:

$$\int \tilde{K}(\mathbf{k}) P(\mathbf{k}) e^{i\mathbf{k} \cdot (\mathbf{q}_1 - \mathbf{q}_2)} \frac{d\mathbf{k}}{(2\pi)^d} = \delta^d(\mathbf{q}_1 - \mathbf{q}_2), \quad (21)$$

meaning the Power Spectrum is the inverse of the Fourier space kernel. This is very interesting when considering how the probability density functional of a GRF may look in Fourier space. The first thing we notice is that the Fourier transform of a Gaussian is also a Gaussian. Secondly since we are dealing with a real field, the fourier modes will not only be independent of each other, but also obey the reality condition, $\tilde{f}(\mathbf{k}) = \tilde{f}^*(-\mathbf{k})$. Lastly, we can also easily conclude that, following the cosmological principle, $P(\mathbf{k}) \rightarrow P(|\mathbf{k}|) = P(k)$. Using all this, we can construct the probability density functional for each Fourier mode:

$$p[\tilde{f}] \propto \exp \left(-\frac{1}{2} \int \frac{d\mathbf{k}}{(2\pi)^2} \frac{|\tilde{f}(\mathbf{k})|^2}{P(\mathbf{k})} \right). \quad (22)$$

However, our field is described by a combination of its Fourier Modes, and to fulfil our goal of computationally simulating these, we want to discretise it and use Fast Fourier Transform (FFT) techniques to generate field realisations. For this, we start by describing the probability density functional in Fourier space:

$$p(\tilde{f}(\mathbf{k}_1), \tilde{f}(\mathbf{k}_2), \dots) = \prod_i \frac{1}{\sqrt{2\pi P(\mathbf{k}_i)}} \exp \left[-\frac{|\tilde{f}(\mathbf{k})|^2}{2P(\mathbf{k})} \right]. \quad (23)$$

This gives us an easy recipe to generate realisations of a GRF [26]:

1. Start with just a realisation of white noise, made of normally distributed random numbers ($N(\mu = 0, \sigma^2 = 1)$) in position space, $n_w(\mathbf{q})$.
2. The Fourier modes of this field will also be normally distributed and satisfy the reality condition $\hat{n}_w(\mathbf{q}) = FFT[n_w(\mathbf{q})](\mathbf{k})$
3. Now we can simply rescale the different modes with the power spectrum, thus changing their standard deviation. Inverting this field will return the right GRF:

$$f(\mathbf{q}) = FFT^{-1}[\sqrt{P(\mathbf{k})} \hat{n}_w(\mathbf{q})] = FFT[n_w(\mathbf{q})](\mathbf{k})$$

We now have a clear method to produce unconstrained realisations of Gaussian Random fields, now we want to explore how to constrain them.

3.2 Constrained Gaussian Random Fields

To study specific structures of the cosmic web, we want to impose constraints on the field that satisfy the conditions for their formation. For this, we need to understand the behaviour of constrained GRF before we attempt to generalise the methodology for a non-Gaussian realisation.

However, we are not only interested in imposing constraints on the field, but these will be non-linear constraints, since gravity itself is non-linear. We will now discuss how the Hoffman-Ribak method for imposing linear constraints on a GRF [34] works, before describing a method to generalise this to impose non-linear constraints as well.

3.2.1 Linear Constraints

Starting with a set of linear constraints:

$$\Gamma = \{C_i[f; \mathbf{q}_i] = c_i, i = 1, \dots, M\}, \quad (24)$$

where c_i is just a constant related to the value of the field at a given point, these can be any type of 1:1 relationship with the field, being for example a set of derivatives, a function of the points of the field and more in general just a convolution of the field with a given kernel ($C[f, \mathbf{q}'] = \int K[\mathbf{q}' - \mathbf{q}]f(\mathbf{q})d\mathbf{q}$).

We know that the constrained field must have the same statistical properties as the original GRF, and its probability density function is defined as:

$$p[f|\Gamma] = \frac{p[f, \Gamma]}{p[\Gamma]}. \quad (25)$$

Similarly, since we are dealing with linear constraints, these will follow a multidimensional Gaussian distribution:

$$p[\Gamma] = \frac{\exp[-\frac{1}{2}\Delta\mathbf{C}^T\mathbf{Q}^{-1}\Delta\mathbf{C}]}{\sqrt{(2\pi)^M \det\mathbf{Q}}}. \quad (26)$$

Now using equation 16, we can obtain an overall expression for the constrained GRF probability density functional:

$$p[f|\Gamma] \propto \exp\left[-\frac{1}{2}\iint f(\mathbf{q}_1)K(||\mathbf{q}_1 - \mathbf{q}_2||)f(\mathbf{q}_2)d\mathbf{q}_1d\mathbf{q}_2 - \Delta\mathbf{C}^T\mathbf{Q}^{-1}\Delta\mathbf{C}\right]. \quad (27)$$

From this we can understand that we can divide any realisation of the field into two separate components: a mean field ($\bar{f}_{\bar{\mathbf{c}}}$) and a residual field $\delta f_{\bar{\mathbf{c}}}$, where the mean fields contains the main signature from the constraints, while the residual field encompasses the randomness of that specific realization.

$$f(\mathbf{q}) = \bar{f}_{\bar{\mathbf{c}}}(\mathbf{q}) + \delta f_{\bar{\mathbf{c}}}(\mathbf{q}) \quad (28)$$

Based on this, we can divide our job into two: separately calculate the mean field and the residual field before constructing our realisation. Before doing this, it is helpful to introduce some notation. To define this field, we will rely on different correlation functions: the correlations of our unconstrained field, between the constraints, and between these two:

$$\xi(\mathbf{q}_1, \mathbf{q}_2) = \langle (f(\mathbf{q}_1) - \bar{f}(\mathbf{q}_1))(f(\mathbf{q}_2) - \bar{f}(\mathbf{q}_2)) \rangle = \text{cov}(f(\mathbf{q}_1), f(\mathbf{q}_2)) \quad (29)$$

$$\xi_i(\mathbf{q}) = \langle (f(\mathbf{q}) - \bar{f}(\mathbf{q}))(C_i - \bar{C}_i) \rangle = \text{cov}(f(\mathbf{q}), C_i) \quad (30)$$

$$\xi_{ij} = \langle (C_i - \bar{C}_i)(C_j - \bar{C}_j) \rangle = \text{cov}(C_i, C_j) \quad (31)$$

The mean field is defined as $\bar{f}_{\bar{\mathbf{c}}}(\mathbf{q}) = \langle f(\mathbf{q})|\Gamma \rangle$, where $\mathbf{c}$ is the vector of the constraints we are imposing on the field and $\bar{\mathbf{c}}$ is the average over all the possible values that $\mathbf{c}$ can take such that the constraints C_i are obeyed. So, how can we write the mean field in a form that is easy to calculate? We start by defining an auxiliary field [27]:

$$g(\mathbf{q}) = f(\mathbf{q}) + A(\mathbf{q}) \cdot \mathbf{C} \quad , \quad A_i(\mathbf{q}) = -\xi_i(\mathbf{q})\xi_{ji}^{-1} \quad (32)$$

We now want to describe this field in terms of known values. To this end, we start by identifying its covariance with respect to various known quantities. First, we can see that this is by definition independent of $\mathbf{C}$ [27]:

$$\text{cov}(g(\mathbf{q}), \mathbf{C}) = \text{cov}(f(\mathbf{q}), \mathbf{C}) + \text{cov}(\mathbf{C}, \mathbf{C})A(\mathbf{q}) \quad (33)$$

$$= \sum_k (\text{cov}(f(\mathbf{q}), C_k) - \delta_{ik}(\text{cov}(f(\mathbf{q}), C_i))) \quad (34)$$

$$= \xi_k(\mathbf{q}) - \xi_i(\mathbf{q})\xi_{ij}^{-1}\xi_{jk} = 0 \quad (35)$$

Similarly, we can write the expectation value of the auxiliary field, $\langle g(\mathbf{q}) \rangle = \bar{f}(\mathbf{q}) + A(\mathbf{q}) \cdot \bar{\mathbf{C}}$, where $\bar{\mathbf{C}}$ is the average over the whole constrained space, meaning we are not applying any constraints on the field yet. This gives us a new way to represent the mean-field:

$$\langle f(\mathbf{q}) | \Gamma \rangle = \langle (g(\mathbf{q}) - A(\mathbf{q}) \cdot \mathbf{C}) | \Gamma \rangle \quad (36)$$

$$= \bar{f}(\mathbf{q}) + A(\mathbf{q}) \cdot (\bar{\mathbf{C}} - \mathbf{c}) \quad (37)$$

$$= \bar{f}(\mathbf{q}) + \xi_i(\mathbf{q})\xi_{ij}^{-1}(c_j - \bar{C}_j) \quad (38)$$

Now we can simplify this by assuming the unconstrained field has a zero mean, as we have done before, and that the overall constraints vector obeys the same ($\bar{f}(\mathbf{q}) = 0$, $\bar{\mathbf{C}} = 0$), giving $\langle f(\mathbf{q}) | \Gamma \rangle = \xi_i(\mathbf{q})\xi_{ij}^{-1}(c_j - \bar{C}_j)$. Similarly, we now check the covariance of the constrained field [27]:

$$\text{cov}(f(\mathbf{q}_1), f(\mathbf{q}_2) | \Gamma) = \langle (\bar{g}(\mathbf{q}_1) - A(\mathbf{q}_1) \cdot \mathbf{C})(\bar{g}(\mathbf{q}_2) - A(\mathbf{q}_2) \cdot \mathbf{C}) | \Gamma \rangle \quad (39)$$

$$= \text{cov}(g(\mathbf{q}_1), g(\mathbf{q}_2) | \Gamma) + \text{cov}(A(\mathbf{q}_1) \cdot \mathbf{C}, A(\mathbf{q}_2) \cdot \mathbf{C}) \quad (40)$$

$$- A(\mathbf{q}_1) \cdot \text{cov}(f(\mathbf{q}_2), \mathbf{C} | \Gamma) - A(\mathbf{q}_2) \cdot \text{cov}(f(\mathbf{q}_1), \mathbf{C} | \Gamma) \quad (41)$$

$$= \text{cov}(f(\mathbf{q}_1), f(\mathbf{q}_2)) + \xi_i(\mathbf{q}_1)\xi_{ij}^{-1}\xi_{jk}\xi_{kl}^{-1}\xi_l(\mathbf{q}_2) - 2\xi_i(\mathbf{q}_1)\xi_{ij}^{-1}\xi_j(\mathbf{q}_2) \quad (42)$$

$$= \xi(\mathbf{q}_1, \mathbf{q}_2) - \xi_i(\mathbf{q}_1)\xi_{ij}^{-1}\xi_j(\mathbf{q}_2)$$

Now we know that the field's covariance is independent of the constraints imposed on it. Similarly, we know that $\langle f(\mathbf{q}) | \Gamma \rangle$ is deterministic, completely determined by the constraints, meaning that it has no variance. Thus we can conclude that the variance of the constrained field will always come from the variance of the residual field, $\langle \delta f^2 | \Gamma \rangle = \sigma_0^2 - \xi_i(\mathbf{q}_1)\xi_{ij}^{-1}\xi_j(\mathbf{q}_2)$, which in its turn is also completely independent from the specific value of the constraints imposed. From these, we can very quickly realise a formula to obtain our mean-field:

$$\delta f = f - \bar{f}(\mathbf{q}) + \xi_i(\mathbf{q})\xi_{ij}^{-1}(c_j - \bar{C}_j) = f - \langle f(\mathbf{q}) | \Gamma \rangle \quad (43)$$

3.2.2 Non-Linear Constraints

The formalism explained so far assumes we are imposing linear constraints on our field. However, in the context of caustic conditions, these are non-linear by definition, since they arise from eigenvalues and eigenvectors, which are non-linear relations.

However, these can be included in a special group of non-linear constraints, constraints that are defined in terms of linear functionals of the field. In our case this means that even though the constrained conditions are written in terms of eigenvalues and eigenvectors of the field and their derivatives, these values can be written in terms of the derivatives of the primordial density field, which are linear on the field we are trying to constrain (we discussed this in further detail in Section 4.1). So we are now interested in discussing how our formalism can be extended to this specific type of non-linear constraints [27].

We start by considering the set of linear functionals of the field $C_i[f, \mathbf{q}_i]$ with $i = 1, \dots, M$. This set can define an M -dimensional space of all the values c_i that these functionals can have. A coordinate in this space can be defined as $\mathbf{c} = (c_1, \dots, c_M) \in \mathbb{R}^M$. In this space of constraint values, we can now define a non-linear constraint as:

$$\Gamma = \{\mathcal{C}(\mathbf{c}) = 0\}_{i=1}^N, \quad (44)$$

this represents a set of N non-linear functions $\mathcal{C} : \mathbb{R}^M \rightarrow \mathbb{R}$ with $k = 1, \dots, N$ and $N \leq M$. We can visualise this as defining a $(M - N)$ -dimensional manifold ($\mathcal{M}_\mathcal{C}$) in the overall constraint space such that every vector $\mathbf{c}$ obeys the non-linear constraints we wish to impose on the field:

$$\mathcal{M}_\mathcal{C} = \{\mathbf{c} | \mathcal{C}(\mathbf{c}) = 0, \quad \text{for all } i = 1, \dots, N\}. \quad (45)$$

This definition then allows us to establish a clear definition of the probability density of these non-linear constraints, it is simply the probability of a vector in linear constrained space $\mathbf{c}$ being within the manifold defined by the non-linear constraints:

$$p(\mathbf{c} | \mathbf{c} \in \mathcal{M}_\mathcal{C}) = \frac{p(\mathbf{c})}{\int_{\mathcal{M}_\mathcal{C}} p(\mathbf{c}) d\mathbf{c}}. \quad (46)$$

The whole process of generating constrained realisations of GRFs becomes now a problem of effectively sampling the manifold and finding the constraints that correspond to our conditions.

Most of the assumptions used to generate constraint realisations for linear conditions remain applicable, namely, we can still assume a differentiable field and generate a mean field that depends on the constraints imposed. Starting with the meanfield, we can define it as:

$$\bar{f}_{\mathcal{C}} = \langle f(\mathbf{q}) | \Gamma \rangle = \int_{\mathbf{c} \in \mathcal{M}_{\mathcal{C}}} \bar{f}_{\mathbf{c}}(\mathbf{q}) p(\mathbf{c} | \mathbf{c} \in \mathcal{M}_{\mathcal{C}}) d\mathbf{c} \quad (47)$$

$$= \xi_i(\mathbf{q}) \xi_{ij}^{-1} \int_{\mathbf{c} \in \mathcal{M}_{\mathcal{C}}} (c_j - \bar{C}_j) p(\mathbf{c} | \mathbf{c} \in \mathcal{M}_{\mathcal{C}}) d\mathbf{c} \quad (48)$$

$$= \xi_i(\mathbf{q}) \xi_{ij}^{-1} (\bar{c}_j - \bar{C}_j) = \bar{f}_{\bar{\mathbf{c}}}, \quad (49)$$

where we defined $\bar{\mathbf{c}}$ using the manifold probability distribution, $\bar{\mathbf{c}} = \int_{\mathbf{c} \in \mathcal{M}_{\mathcal{C}}} \mathbf{c} p(\mathbf{c} | \mathbf{c} \in \mathcal{M}_{\mathcal{C}}) d\mathbf{c} = \langle \mathbf{c} | \Gamma \rangle$.

However, one property changes: the residual field, δf , is no longer constraint-independent and is no longer a GRF. This can be seen by directly studying its variance [27]:

$$\langle \delta f(\mathbf{q})^2 | \Gamma \rangle = \int_{\mathbf{c} \in \mathcal{M}_{\bar{\mathcal{C}}}} \langle (f(\mathbf{q}) - \bar{f}_{\bar{\mathbf{c}}}(\mathbf{q}))^2 | C_i[f, \mathbf{q}_i] = c_i \rangle p(\mathbf{c} | \mathbf{c} \in \mathcal{M}_{\bar{\mathcal{C}}}) d\mathbf{c} \quad (50)$$

$$= \int_{\mathbf{c} \in \mathcal{M}_{\bar{\mathcal{C}}}} \langle (f(\mathbf{q}))^2 | C_i[f, \mathbf{q}_i] = c_i \rangle p(\mathbf{c} | \mathbf{c} \in \mathcal{M}_{\bar{\mathcal{C}}}) d\mathbf{c} - (\bar{f}_{\bar{\mathbf{c}}}(\mathbf{q}))^2 \quad (51)$$

$$= \sigma_0^2 - \xi_i(\mathbf{q}) \xi_{ij}^{-1} \xi_j(\mathbf{q}) + \sum_{i,j,k,l=1}^M \xi_i(\mathbf{q}) \xi_{ij}^{-1} \text{cov}(c_j, c_k | \mathbf{c} \in \mathcal{M}_{\bar{\mathcal{C}}}) \xi_{kl}^{-1} \xi_l(\mathbf{q}) \quad (52)$$

$$= \sigma_0^2 - \sum_{i,j=1}^M \xi_i(\mathbf{q}) \zeta_{ij}^{-1} \xi_j(\mathbf{q}), \quad (53)$$

where we defined the $\zeta_{ij}^{-1} = \xi_{ij}^{-1} - \sum_{k,l=1}^M \xi_{ik}^{-1} \text{cov}(c_j, c_k | \mathbf{c} \in \mathcal{M}_{\bar{\mathcal{C}}}) \xi_{jl}^{-1}$. However, this won't be too problematic, since as we have shown in equation 43, the residual field can be found by subtracting the mean-field of an unconstrained realisation (the mean field defined by the natural constraints that our GRF has, before we impose any upon it) from the original unconstrained GRF. Thus, we can easily generate an algorithm capable of obtaining the residual field, even for non-linear constraints

3.2.3 Generating a Constrained GRF

The nature of linear constraints works very well with the algorithm we use to generate GRF. This method is known as the Hoffman Ribak algorithm [35]. Using fast Fourier transforms, we can quickly calculate the value of a constraint C_i using the convolution theorem evaluated at the origin:

$$C_i[f, \mathbf{q}_i] = \text{FFT}^{-1}[\tilde{C}_i^* \tilde{f}(k)](\mathbf{0}). \quad (54)$$

Where we define $\tilde{C}_i^*$ as the Fourier transform of the operator acting on the field. Similarly, we also know, from the method we use to generate a GRF, that $\tilde{f}(k) \propto \sqrt{P(\mathbf{k})}$. Following this gives us an easy way to now generate the correlation functions needed to describe a constrained field:

$$\xi_i(\mathbf{q}) = \text{FFT}^{-1}[\tilde{C}_i^*(\mathbf{k})P(\mathbf{k})](\mathbf{q}), \quad (55)$$

$$\xi_{ij} = \text{FFT}^{-1}[\tilde{C}_i^*(\mathbf{k})P(\mathbf{k})\tilde{C}_j(\mathbf{k})](\mathbf{0}), \quad (56)$$

giving us now all we need to reliably generate constrained realisations of a constrained field integrated into our previous method to generate GRFs:

1. Start by generating a GRF (f) with the desired properties.
2. Evaluate the natural constraints of the field (n_i), and use these to obtain the unconstrained mean field, and thus get the residual field ($\delta f = f - \xi_i(\mathbf{q})\xi_{ij}n_j = \delta\Phi$).
3. Separately evaluate the constraints (c_j) that we wish to impose on the field, and use them to calculate the desired mean field ($\langle\Phi(\mathbf{q})|\Gamma\rangle = \xi_i(\mathbf{q})\xi_{ij}c_j$).
4. Put these ingredients together to generate a new constrained field with the desired constraints and residual field ($\Phi = \langle\Phi(\mathbf{q})|\Gamma\rangle + \delta\Phi = \xi_i(\mathbf{q})\xi_{ij}c_j + f - \xi_i(\mathbf{q})\xi_{ij}n_j$).

This method relies on the fact that our field can always be decomposed into a mean field, constrained by the imposed constraints, and a residual field, which is constrained independently. The main difference between applying this method to linear and non-linear constraints is the sampling method used to obtain the constraints. In the non-linear case, step 3 will involve using a sampler capable of providing different realisations of constraints that satisfy the properties we want to impose on the field, whereas this is not necessary for the linear case.

This allows us to generate a field with any constraints we might desire. These can be the mean constraints, $\bar{\mathbf{c}}$, or can be any specific constraint vector that obeys the constraints we set up, $\mathbf{c}$. This distinction is important because, as we use progressively more complex constraints, the fact that the sampled manifold represents a non-linear space becomes more relevant, once, since these have a different geometry, the euclidean average of the points of the manifold will not necessarily be contained within the manifold, and thus $\bar{\mathbf{c}}$ might not always be a valid constraint. As such, being able to study several $\mathbf{c}$ configurations when $\bar{\mathbf{c}}$ is not available is an important approach we must take if we wish to study the properties of the set of constrained realisations for a non-linear constraint manifold.

4 Simulating the Cosmic Web

In section 2 we explained how we can study structures in the cosmic web using only the properties of the primordial gravitational potential, whereas in section 3 we have shown how we can not only generate different realizations of this potential, but also manipulate them in a way they obey specific constraints.

Figure 5 shows the workflow of how we implement the methods previously described to study specific structures of the cosmic web. First, we define and sample the constraints we want to impose, based on the structure we are interested in studying. Simultaneously, based on its power spectrum, we generate an unconstrained realisation of a GRF. Using these two together, we can generate a constrained GRF, compute the eigenvalue fields, and subsequently use them to identify the underlying structure via the caustic conditions. In this section, we aim to provide a practical example of how this method works for an A_3 caustic. We will start by explaining how we can obtain our constraints and sample them, before discussing how to generate a realistic power spectrum in order to show the final field for a 2D example.

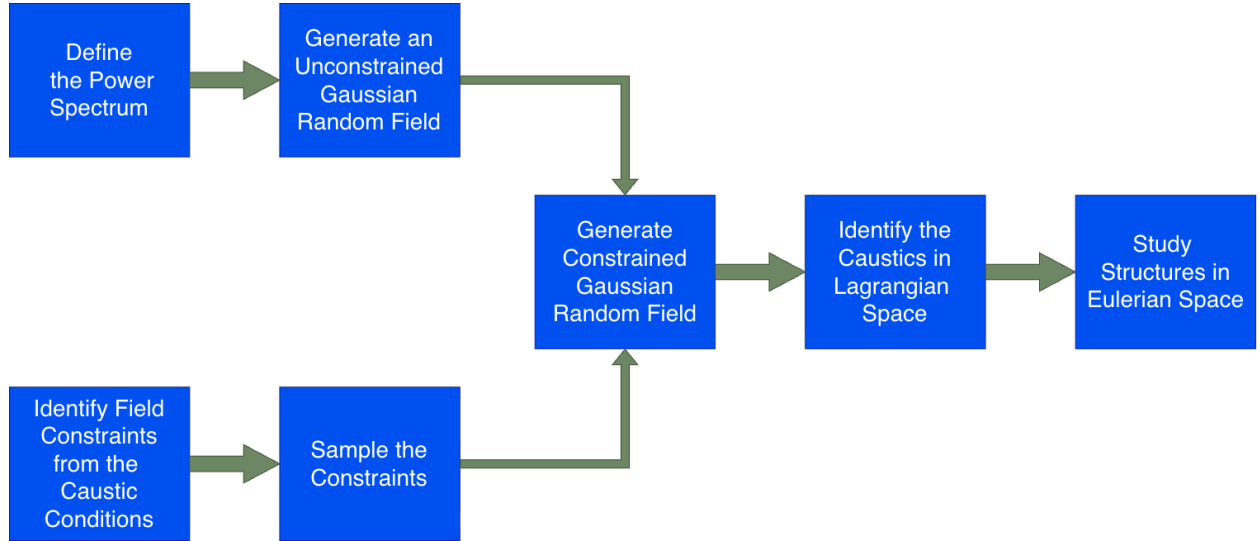


Figure 5. How to simulate the Cosmic Web. The following diagram presents an overview of our method for studying specific structures in the cosmic web. By identifying the constraints a field must obey to generate a different caustic, we can use the Hoffman-Ribak method described in section 3.2 to generate realisations of specific structures of the cosmic web and study their properties.

4.1 Constrained A_3 conditions for a GRF

Now that we have developed the machinery needed to generate constrained Gaussian Random Fields, it is important to explain how we can apply this to actual structures of the cosmic web. In this report, we are interested in studying constrained A_3 type of structures, 2D filaments

and 3D cosmic walls. Here, we present an overview of how these can be constrained for a GRF. This will provide the basis for our end goal of generalising this method to non-Gaussian random fields, which we will discuss in Section 5.

First, we start with the conditions for the A_3 cusp caustic. From the caustic conditions, we know that these can be completely determined in terms of the derivatives of the deformation tensor. We can write this in terms of derivatives of the displacement field:

$$M_{ij} = -D_+(t) \frac{\partial^2 \psi(\mathbf{q})}{\partial q_i \partial q_j}, \quad (57)$$

meaning we can now define our derivatives in terms of the eigenvalues (λ_i) of the time-independent deformation tensor, $\frac{\partial^2 \psi(\mathbf{q})}{\partial q_i \partial q_j}$, such that $\mu_{i,t} = -D_+(t)\lambda_i$. From equation 15, we can rewrite our equations considering this simplification as well as considering these conditions in terms of the first eigenvector, the biggest eigenvector:

$$\lambda_1 = 1/D_+(t) \quad \text{and} \quad \mathbf{v}_1 \cdot \nabla \lambda_1(\mathbf{q}) = 0 \quad (58)$$

Adding to these two cusp conditions, we are also interested in studying specifically the centre of the cusp. This will lead to an additional set of constraints that depend on whether we are in 2 or 3 dimensions, as explained in [28] and [29].

4.1.1 2D constraints

At the centre of a cusp caustic, the mass flow must be maximum. This is because mass will flow towards denser regions, represented by higher-order caustics at the end of the filament. As such, we must impose the maximum stretch condition. We will only state these since their derivation is outside the scope of this project, for a full derivation, please refer to [28, 29]. In 2D, this condition is defined as:

$$\mathbf{v}_2 \cdot \nabla \lambda_2 = 0, \quad (59)$$

where λ_2 is the second eigenvalue of the displacement field such that $\lambda_1 \geq \lambda_2$. Joining all the conditions in, we now have a full map of the constraints that we must impose:

2D Centre of a Filament conditions:

$$\lambda_1 = 1/D_+(t) \quad , \quad \mathbf{v}_1 \cdot \nabla \lambda_1(\mathbf{q}) = 0 \quad , \quad \mathbf{v}_2 \cdot \nabla \lambda_2 = 0 \quad \text{and} \quad \lambda_1 \geq \lambda_2 \quad (60)$$

But how can we impose these constraints on the field? As we have been mentioning, we are now defining the caustic conditions not in terms of an abstract deformation tensor, but in terms of the deformation tensor of the displacement field (ψ). From equation 6, this field

can be described in terms of the primordial gravitational potential (ϕ_0). It is this field that we simulate when generating our initial GRF, and it is this field that we must constrain in order to observe a filament. So we need to write the constraints described in equation 60 in terms of linear operators of ψ .

We can then write the time-independent deformation tensor explicitly in terms of derivatives of ψ :

$$M = \begin{pmatrix} \frac{\partial^2 \psi(\mathbf{q})}{\partial q_1 \partial q_1} & \frac{\partial^2 \psi(\mathbf{q})}{\partial q_1 \partial q_2} \\ \frac{\partial^2 \psi(\mathbf{q})}{\partial q_1 \partial q_2} & \frac{\partial^2 \psi(\mathbf{q})}{\partial q_2 \partial q_2} \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} \\ T_{12} & T_{22} \end{pmatrix}, \quad (61)$$

where we have abbreviated the derivatives of the field as $T_{ijk\dots} = \left(\frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} \frac{\partial}{\partial x_k} \dots \right) \psi$, where every index has the value of 1 or 2 to represent one of the spatial derivatives of the field. From this, we can write each eigenvalue and eigenvector in terms of these quantities and thus turn our centre of filament conditions into constraints on the manifold of the derivatives of the displacement field.

Furthermore, we can, without loss of generality, always work in the eigenframe ($T_{12} = 0$) and define $T_{11} \geq T_{22}$. This gives us a straightforward way to obtain the values for the eigenvalues and the eigenvectors:

$$\begin{aligned} \lambda_1 &= T_{11} \quad , \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \lambda_2 &= T_{22} \quad , \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned} \quad (62)$$

However, if we are interested in higher-order derivatives of the eigenvector and eigenvalue fields (which we need for the conditions for our A_3 caustics), these are non-linear relations, and as such, we must solve them first before imposing further conditions such as the eigenframe condition. The method for finding these definitions is to express the deformation tensor in terms of its invariants. To do this, we start by describing the deformation tensor in terms of its characteristic polynomial:

$$\chi(\lambda) = \det(M - \lambda I) = \prod_i (\lambda - \lambda_i), \quad (63)$$

where I is the identity matrix and λ_i is one of the eigenvalues of the matrix. For our tensor, we can get the result for this polynomial for different powers of λ :

$$\chi(\lambda) = \lambda^2 - (T_{11} + T_{22})\lambda + (T_{11}T_{22} - T_{12}^2) = \lambda^2 - \text{Tr}(M)\lambda + \det(M). \quad (64)$$

This equation is the same for any frame, meaning that the determinant and the trace of the matrix are independent of the frame in which we describe the matrix. We call these the matrix invariants. We can then use these invariants to establish a relation between the eigenvalues and the matrix elements in any frame. To understand why this is relevant, we can consider the concrete example of the invariants of the deformation tensor:

$$\text{Tr}(M) = \lambda_1 + \lambda_2 = T_{11} + T_{22} , \quad (65)$$

$$\det(M) = \lambda_1 \lambda_2 = T_{11} T_{22} - T_{12}^2 . \quad (66)$$

If we want to find the first- and second-order derivatives of our eigenvalues, we can take the spatial derivatives of these invariants and solve the resulting system of equations to obtain the required values. Only once we have calculated these relations do we apply the eigenframe condition. Using this method, one can capture the non-linear behaviour of the eigenvalues that we would have lost if we had started simply with the diagonalised matrix.

An example for this is the case of $\partial_x \lambda_1$. If we got this simply by considering the diagonal matrix, we would get it as being T_{1111} , which is wrong since the true value of this quantity is $\frac{2T_{112}^2}{T_{11}-T_{22}} + T_{1111}$.

The same method can be applied for the derivatives of the eigenvectors, using the eigenvector equations from:

$$\begin{pmatrix} T_{11} - \lambda_1 & T_{12} \\ T_{12} & T_{22} - \lambda_i \end{pmatrix} \begin{pmatrix} v_{ix} \\ v_{iy} \end{pmatrix} = \lambda_i \begin{pmatrix} v_{ix} \\ v_{iy} \end{pmatrix} \quad (67)$$

Using this method, we get the following conditions for the constraint cusp in two dimensions:

$$T_{11} = \frac{1}{D_+(t)} , \quad T_{12} = 0 , \quad T_{22} \leq \frac{1}{D_+(t)} , \quad T_{111} = 0 \quad \text{and} \quad T_{222} = 0 \quad (68)$$

Any point that corresponds to these conditions in the field ψ will thus correspond to the centre of a filament. Now we want to sample the space of derivatives of ψ such that these conditions are obeyed. However, we are missing one detail. While these conditions allow us to impose the existence of the centre of a 2D cusp caustic at a single point of the cosmic web (in our case, we choose this to be the centre of our simulation), we are not controlling the orientation of the cusp. This will be problematic, since we are interested in studying the entire structure rather than a single point. To correct for this, we must also impose an orientation on the cusp.

The normal vector of a cusp is determined as:

$$\mathbf{n} = \nabla (\mathbf{v}_1 \cdot \nabla \lambda_1) = \begin{pmatrix} \frac{3T_{112}^2}{T_{11}-T_{22}} + T_{1111} \\ \frac{3T_{112}T_{122}}{T_{11}-T_{22}} + T_{1112} \end{pmatrix} \quad (69)$$

From this, we can find the angle of the cusp orientation with the $\mathbf{v}_1$ vector:

$$\alpha = \text{sign}(\mathbf{v}_2 \cdot \mathbf{n}) \arccos\left(\frac{|\mathbf{v}_1 \cdot \mathbf{n}|}{|\mathbf{n}|}\right) \quad (70)$$

What we want is to impose the same orientation on all constrained realisations of the cusp so that, when we calculate the average constrained field or study other statistical properties of the sampled realisations, we always consider a cusp with the same orientation. For this, we shift the orientation of the constraints themselves before imposing them on the field, rotating them in such a way that the final cusp orientation always corresponds to the vertical direction ($\alpha = 0$).

4.1.2 3D constraints

In 3D, the main difference is that we have an extra eigenvalue and eigenvector, meaning the centre of a cosmic wall conditions must change to account for this, as described in [29]. This time we want to ensure maximum stretch on both the $\mathbf{v}_2$ and $\mathbf{v}_3$ directions, which can be defined as:

$$\mathbf{v}_2 \cdot \nabla(\lambda_2 + \lambda_3) = 0 \quad \text{and} \quad \mathbf{v}_3 \cdot \nabla(\lambda_2 + \lambda_3) = 0 \quad (71)$$

Joining this definition, with the conditions we previously discussed for the existence of an A_3 caustic, we get the following conditions:

3D Centre of a Wall conditions:

$$\begin{aligned} \lambda_1 &= 1/D_+(t) \quad , \quad \mathbf{v}_1 \cdot \nabla \lambda_1(\mathbf{q}) = 0 \\ \mathbf{v}_2 \cdot \nabla(\lambda_2 + \lambda_3) &= 0 \quad , \quad \mathbf{v}_3 \cdot \nabla(\lambda_2 + \lambda_3) = 0 \quad \text{and} \quad \lambda_1 \geq \lambda_2 \geq \lambda_3 \end{aligned} \quad (72)$$

To find how to express these conditions in terms of the derivatives of the displacement field, we use the same process as the two-dimensional case. In three dimensions, the time-independent deformation tensor is defined as:

$$M = \begin{pmatrix} T_{11} & T_{12} & T_{13} \\ T_{12} & T_{22} & T_{23} \\ T_{13} & T_{23} & T_{33} \end{pmatrix} \quad (73)$$

meaning the characteristic polynomial is described as:

$$\chi(\lambda) = -\lambda^3 + \text{Tr}(M)\lambda^2 - \frac{1}{2}((\text{Tr}(M))^2 - \text{Tr}(M^2))\lambda + \det(M). \quad (74)$$

The existence of 3 dimensions now implies the existence of 3 matrix invariants:

$$\text{Tr}(M) = \lambda_1 + \lambda_2 + \lambda_3 = T_{11} + T_{22} + T_{33}, \quad (75)$$

$$\begin{aligned} \frac{1}{2} ((\text{Tr}(M))^2 - \text{Tr}(M^2)) &= \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 \\ &= T_{11}T_{22} + T_{11}T_{33} + T_{22}T_{33} - T_{12}^2 - T_{13}^2 - T_{23}^2, \end{aligned} \quad (76)$$

$$\det(M) = \lambda_1\lambda_2\lambda_3 = T_{11}T_{22}T_{33} + 2T_{12}T_{13}T_{23} - T_{11}T_{23}^2 - T_{22}T_{13}^2 - T_{33}T_{12}^2. \quad (77)$$

The process to find the constrained conditions from this is then the same, with the exception that we now have more eigenframe conditions: $T_{12} = 0$, $T_{13} = 0$ and $T_{23} = 0$. The same is true for the eigenvector conditions with an extra eigenvector $\mathbf{v}_3 = (0 \ 0 \ 1)$. Following this procedure, we get the following constrained conditions for a 3-dimensional cosmic wall:

$$\begin{aligned} T_{11} &= \frac{1}{D_+(t)} \quad , \quad T_{12} = 0 \quad , \quad T_{13} = 0 \quad , \quad T_{22} \leq \frac{1}{D_+(t)} \quad , \quad T_{23} = 0, \\ T_{111} &= 0 \quad , \quad T_{222} = -T_{233} \quad \text{and} \quad T_{223} = -T_{333} \end{aligned} \quad (78)$$

We won't be producing simulations for the three-dimensional constrained fields due to time limitations, focusing instead on demonstrating how the constraints in this case work and on using the 2D simulations to validate them. However, it is also relevant to note how we can define the wall's orientation, as in the previous case. In 3 dimensions, we will need two vectors to fully identify the direction of the wall:

$$\mathbf{n}_c = \nabla(\mathbf{v}_1 \cdot \nabla\lambda_1) \quad (79)$$

$$\mathbf{n}_f = \nabla\lambda_1 \quad (80)$$

where $\mathbf{n}_c$ represents the orientation of the cusp sheet and $\mathbf{n}_f$ the orientation of the multistream region where the wall is found. We can see this as the fact that we are in three dimensions, adding an extra rotational symmetry and thus forcing us to consider the two underlying elements that form a cusp in order to break the object's extra symmetry. These can also be expressed in terms of the field's derivatives, as before, using the same method.

4.2 Sampling the Constraints

Having shown how we can generate constraints for both a 2D and a 3D field, we now want to find an effective way to sample the constraint manifold so that we can generate an average constrained field that corresponds to the properties of this space. First, we start by defining the space we are interested in, the linear derivative space, which can be defined in 2D as:

$$\mathbf{T} = \{T_{11}, T_{12}, T_{22}, T_{111}, T_{112}, T_{122}, T_{222}, T_{1111}, T_{1112}, T_{1122}, T_{1222}, T_{2222}\} \quad (81)$$

$$= \{\mathbf{T}^{(2)}, \mathbf{T}^{(3)}, \mathbf{T}^{(4)}\}, \quad (82)$$

where $\mathbf{T}^{(2)}$ is the space of second order derivatives, $\mathbf{T}^{(3)}$ is the space of third order derivatives and $\mathbf{T}^{(4)}$ is the space of fourth order derivatives. We can use the same definition for a 3D space, the only difference is that each space must now account for the existence of the third spatial dimension. This derivative space is linear and, as such follows a multidimensional Gaussian distribution:

$$p(\mathbf{T})d\mathbf{T} \propto \exp\left(-\frac{1}{2}\mathbf{T}\Sigma\mathbf{T}^{-1}\right)d\mathbf{T} \quad (83)$$

where Σ is the covariance matrix, which can be written in the following block form :

$$\Sigma = \Sigma^{(2,3,4)} = \begin{pmatrix} \Sigma_{22} & 0 & \Sigma_{24} \\ 0 & \Sigma_{33} & 0 \\ \Sigma_{24}^T & 0 & \Sigma_{44} \end{pmatrix}, \quad (84)$$

where Σ_{ij} is the correlation matrix between the i^{th} and j^{th} order derivative spaces. The values of the individual elements of these matrices must be calculated directly from the expression for $\langle T_{ijk...}T_{lmn...} \rangle$ for either two or three dimensions. Full calculations of these have been shown in [28] and [29], while these are essential to our methods, they are not especially clarifying, and as such, we will just quote the results from these papers³.

Now we can consider what imposing the caustic conditions will do to this derivative space. First, we must consider what it means to impose our eigenframe condition; this is the equivalent of rotating the derivative space into a new space. This means the differential measure $d\mathbf{T}$ will change in this new frame, by a factor determined by the Hessian of this rotation. This change in 2D is given by $d\mathbf{T} \rightarrow (T_{11} - T_{22})d\mathbf{T}$, as shown in [27]. From this, we can get the new probability distribution: [36]:

³We also show a brief derivation of how to get the 2D matrices in Appendix B

$$p(\mathbf{T})d\mathbf{T} \propto \exp\left(-\frac{1}{2}\mathbf{T}\Sigma\mathbf{T}^{-1}\right)(T_{11} - T_{22})d\mathbf{T} \quad (85)$$

$$\propto \exp\left(-\frac{1}{2}\mathbf{T}\Sigma\mathbf{T}^{-1} + \ln(T_{11} - T_{22})\right). \quad (86)$$

This can be seen as a generalisation of the Doroshkevich formula for the distribution of eigenvalues [36], and shows clearly how the change to the eigenframe makes the distribution non-Gaussian. In 3D the metric will be given by $d\mathbf{T} \rightarrow (T_{11} - T_{22})(T_{11} - T_{33})(T_{22} - T_{33})d\mathbf{T}$, and the same principle would apply, as seen in [29].

Now that our probability distribution is in the eigenframe, we just need to set a vector $\mathbf{Y}$ with our constraints, thus introducing a further non-linear term in the sampled distribution. The 2D case is described using the constraints shown in Equation 68:

$$\mathbf{Y} = \left\{ \frac{1}{D_+(t)}, 0, T_{22}, 0, T_{112}, T_{122}, 0, T_{1111}, T_{1112}, T_{1122}, T_{1222}, T_{2222} \right\}. \quad (87)$$

We can then sample the different realisations of $\mathbf{Y}$ using the probability $p(\mathbf{Y})$, as described in Equation 86. For this, we can use different sampling methods. However, this probability distribution seems uniquely suited for a Hamiltonian Monte Carlo (HMC) sampler [37, 38].

A Hamiltonian Monte Carlo sampler works by turning the initial probability distribution into a potential, and then treating the sample points as objects in orbit about that potential. This way, it will always follow the points in the direction of increasing probability, thus reproducing the correct probability distribution with a very low rejection rate. In our 2D case, the sampler would be studying a potential given by:

$$U(\mathbf{Y}) = -\ln(p(\mathbf{Y})) = \frac{1}{2}\mathbf{Y}\Sigma\mathbf{Y}^{-1} - \ln\left(\frac{1}{D_+(t)} - T_{22}\right), \quad (88)$$

where we can ignore the normalisation factor of the probability distribution, since it is constant and only shifts the origin of the potential. Using an HMC sampler with this probability distribution, we can now generate several realisations of the vector $\mathbf{Y}$ that obey our constraint conditions, representing different instances of the constrained field.

However, before using this to calculate the average constraints, we must set the correct orientation. We measure it using the normal vectors defined earlier to determine the orientation of the cusp point, then rotate each constraint value to align with the first eigenvector. This can be easily done, once we have the orientations, because each family of derivatives exists on its own linear space that can be rotated individually:

$$\mathbf{T}_{\text{rotated}}^{(i)} = (R^{(i)}(-\theta))^T \mathbf{T}^{(i)} R^{(i)}(-\theta), \quad (89)$$

where θ is the angle defined by the orientation with the first eigenvector and $R^{(i)}$ is the rotation matrix, defined by the group symmetry these derivative spaces obey, SO(2) for the 2D case and SO(3) for the 3D case. Each rotation matrix is an n -dimensional representation of the underlying symmetry, where n refers to the size of the family of derivatives (for example, in 2D, $n = 3$ for $T^{(2)}$)⁴.

Once we have obtained the correct rotated constraints, we average them to obtain the average constraint vector. Using the Hoffman-Ribak method described in section 3.2, we can impose these constraints in an initial GRF to force it to generate a filament at its centre. But before this, we need to generate our unconstrained primordial displacement field.

4.3 Generating the Initial GRF

The properties of the later universe cosmic structure are encoded in the primordial gravitational potential (ϕ_0), which can be used to define the displacement field $\psi(\mathbf{q}) = \frac{2}{3\Omega_0 H_0^2} \phi_0(\mathbf{q})$. In this case, we consider that ϕ_0 is a GRF, and as such, all its properties are contained in its power spectrum. For this study, we consider a Harrison-Zel'dovich scale invariant power spectrum [28, 39, 40], of the form:

$$P(k) = N k^{n-4} e^{-R_s^2 k^2}, \quad (90)$$

where R_s is the cut-off length, smoothing off the field for large wavelengths, and N is a normalisation constant given by:

$$N = \frac{\alpha^2 4\pi R_s^{n-2}}{\Gamma(1 + \frac{n}{2})}. \quad (91)$$

Where we define $n = 1$ in line with the recent Planck data of $n_s = 0.965 \pm 0.004$ [5], thus giving us an overall form $P(k) \propto k^{-3}$, and use a cut-off length $R_s = 1$ to ensure the field varies smoothly at short distances. Adding to this, we set $\alpha = \sqrt{10}$, where α is the field amplitude and can be seen as setting the time scale of the simulation[28]. Using the algorithm described in Section 3.1, we can generate a realisation of ϕ_0 that obeys the properties defined by this power spectrum. Once we have a full representation of ϕ_0 , we use the values of $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, in line with current measurements [41] and assume a flat universe, $\Omega_0 = 1$, to obtain the field ψ that we use for the study of the caustic structure.

4.4 Identifying the caustic structure

Lastly, we want to join all we have been discussing to generate realisations of the cosmic web with its caustic structure overlaid. Figure 6 shows the results of implementing this method

⁴Appendix C shows a derivation of these matrices for the 2D case, while the 3D case can be found in [29].

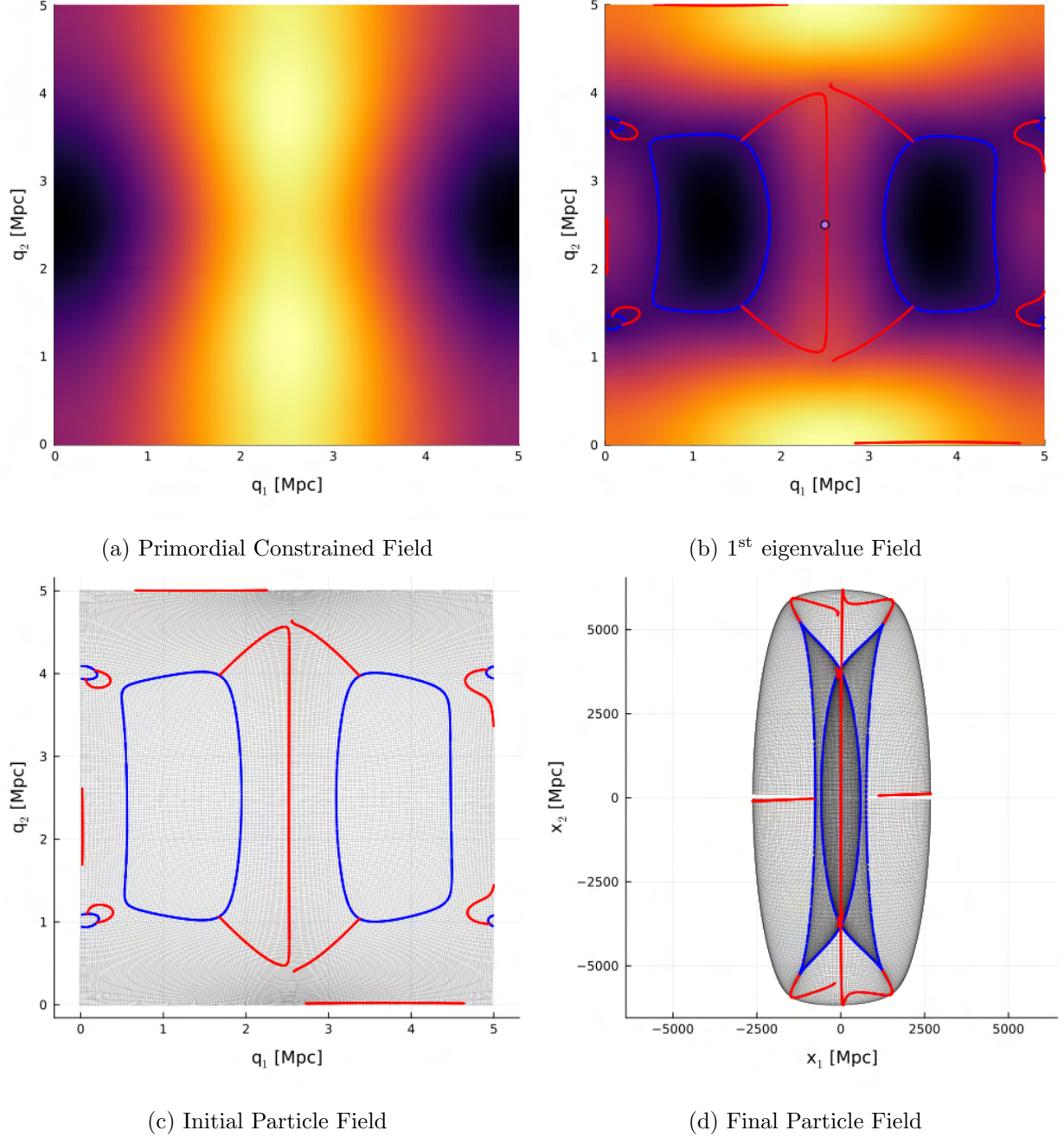


Figure 6. 2D constrained filament, here we see how the generated constrained Gaussian random field reproduces a filament in the cosmic web ($\mathbf{q}$ represents the initial Lagrangian space, while $\mathbf{x}$ represents the final Eulerian space). Here, the blue lines represent the A_2 caustics while the A_3 lines are represented in red. We can see a clearly defined filament at the centre of the field.

for a 2D realisation. We start by using the conditions we set in section 4.1, together with the sampling method described in section 4.2 to find the constraint values we want to impose on the primordial GRF. In this case, we choose to apply the mean-constraints from our sampling, since these still represent a cusp. To obtain these, we used our own HMC sampler,

assuming $m = 1$ for the test mass, with an integration time step of 1 and performing 100 time steps for each of the 1000 samples taken.

Using these values, together with Equations 55 and 56, we apply the method described in section 3.2.3 to find the constrained GRF, seen in Figure 6a. This is generated based on an initial unconstrained GRF corresponding to the power in Equation 90, generated using the method described in Section 3.1 for a 256×256 grid.

Once we have obtained the constrained GRF, we generate its first eigenvalue, the space where the caustic conditions live, as seen in Figure 6b. From this field, we can identify the points where our different conditions are met. In our case, we use a marching-triangle algorithm, dividing our grid into triangles and identifying the sites on either side of a caustic point, thereby allowing us to extrapolate the positions of different caustics. This allows us to identify both the A_3 caustic lines indicating a filament, as well as the A_2 lines representing the bounds of multistream regions around A_3 lines.

From here, we can juxtapose these points within the initial particle field, the field corresponding to the mass distribution of the early universe, as seen in Figure 6c. Evolving these two structures (the field and the caustic points) using the Zel'dovich approximation, we can then observe the late universe filament formed, surrounded by a multistream region, whose boundaries are defined by an A_2 line, represented in Figure 6d.

5 Non-Gaussianities

The method described in section 4 summarises how we can use Gaussian Random Fields to simulate conditions in the early universe and study the evolution of its structures. However, many early-universe theories predict that the primordial gravitational potential may exhibit a non-Gaussian behaviour. In this section, we aim to present, for the first time, a study of how the methodology described so far can be extended to address non-Gaussian random fields and to examine how structures in the cosmic web will evolve under their influence.

Figure 7 shows how we aim to modify our method to account for non-Gaussian Random Fields. We need to find a way to define non-Gaussian fields as a construct of a GRF, and describe their constraints in terms of linear operators of a GRF, so that we can use the Hoffman-Ribak method to constrain realisations of the GRF that will be used to construct the non-Gaussian field.

In this section, we aim to do just that. We will start by describing how this method can be extended to the simplest type of non-Gaussian fields, the Local type non-Gaussianities, and show how the constraints for an A_3 caustic in these fields look. Secondly, we will introduce the formalism that allows us to describe a generic non-Gaussianity, explain why this makes it incompatible with the Hoffman-Ribak algorithm, and present two alternative methods we can use to find constraints for a generic non-Gaussian field within the formalism we have described so far, using approximations of the field.

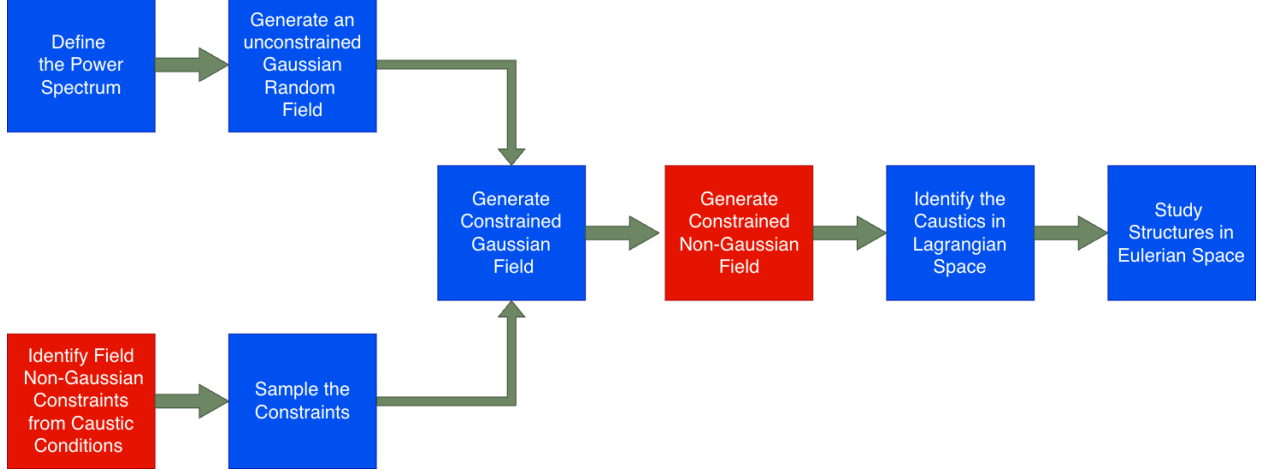


Figure 7. Implementing non-Gaussianities. The following diagram shows how the workflow used to generate constrained realisations of Gaussian random fields and to study the different structures of the Cosmic Web needs to be changed in order to account for non-Gaussianities.

5.1 Local Non-Gaussianities

Local non-Gaussian fields are the simplest example of a non-Gaussian field we can observe, they are predicted by different early universe theories such as the curvaton model or the varying decay rate inflation model [8, 42, 43]. These are an exception within the non-Gaussian fields due to their local nature, since this means they can be described in terms of local operators of the field, in this case, multiples of the field itself, providing us with an easy way to describe these fields without the need to use the higher order statistics to define their properties. As such, a Local non-Gaussian field can be described by the following expression:

$$\Phi_{Local} = \phi + f_{NL} (\phi^2 - \langle \phi^2 \rangle) , \quad (92)$$

where ϕ is the initial GRF we have generated and f_{NL} is a free parameter that parameterises the amplitude of the non-Gaussianities. We can set $\langle \phi^2 \rangle = 0$, as we have been doing for the Gaussian case.

We now have a clear way to generate a Local non-Gaussian field based on a GRF realisation. The challenge now is to find a method to generate constraints for this initial GRF such that, when combined to form this new field, we obtain the desired structure we want to study. For this, we need to adapt the method described in section 4.1 and expand it for this field.

The first thing to notice is that the A_3 conditions for this field remain the same, the only thing that changes is the deformation tensor for which we define them, in this case our deformation tensor is now defined in terms of the derivatives of the non-Gaussian Local Field, Φ_{Local} , from here we can try and construct the constraints imposed on this field in terms of constraints of the initial GRF ϕ by explicitly writing the different dependencies. For simplicity, we will assume from here onwards that the fields' definitions include the $\frac{2}{3\Omega_0 H_0^2}$ factor in their definition, and thus we will treat them as displacement fields.

5.1.1 2D constraints

To obtain a constrained cusp in 2D, we must obey the same conditions as presented in equation 60. However, this time we must define the deformation tensor not in terms of the derivatives of the GRF but in terms of the derivatives of the Local non-Gaussian field Φ_{Local} :

$$M = \begin{pmatrix} \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_1} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_2} \\ \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_2} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_2 \partial x_2} \end{pmatrix}. \quad (93)$$

We can write each of these terms in terms of ϕ , for example:

$$\frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_1} = 2f_{NL} (T T_{11} + T_1^2) + T_{11}, \quad (94)$$

where T is the value of the field ϕ at the centre of the box where we are fixing our cusp point, and we keep using the $T_{ijk\dots}$ notation to represent the spatial derivatives of the GRF.

From this, we can see how we can still apply the same field invariants-based method to identify the constraints for a non-Gaussian Local field in terms of the derivatives of the initial GRF. This fact is an exception, in general, non-Gaussian fields will require other types of constraints. It is the local nature of this field that allows for the dependence on the derivatives alone, since these types of constraints are local by nature as well.

Following the procedure outlined here, we get the following constraints for a 2D filament:

Local Constrained Filament Conditions:

A₃ conditions ($\lambda_1 = 1/D_+(t)$, $\mathbf{v}_1 \cdot \nabla \lambda_1(\mathbf{q}) = 0$):

$$T_{11} = -\frac{2D_+(t)f_{NL}T_1^2 - 1}{D_+(t)(2f_{NL}T + 1)}, \quad (95)$$

$$T_{111} = \frac{6f_{NL}T_1(2D_+(t)f_{NL}T_1^2 - 1)}{D_+(t)(2f_{NL}T + 1)^2}. \quad (96)$$

Maximum Stretch Condition ($\mathbf{v}_2 \cdot \nabla \lambda_2 = 0$):

$$T_{222} = -\frac{6f_{NL}T_2T_{22}}{2f_{NL}T + 1}. \quad (97)$$

Eigenframe Condition ($\frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_2}$):

$$T_{12} = -\frac{2f_{NL}T_1T_2}{2f_{NL}T + 1}. \quad (98)$$

First Eigenvalue Condition ($\lambda_1 > \lambda_2$):

$$2f_{NL}(TT_{22} + T_2^2) + T_{22} \leq \frac{1}{D_+(t)}. \quad (99)$$

Where these constraints can be checked to reduce to the Gaussian case for $f_{NL} = 0$. We can see from here that we now have a larger constraint space to consider, since we include both zeroth- and first-order derivatives (T , T_1 , and T_2). Meaning that while our sampling method does not change with the inclusion of Local non-Gaussianities, we need to include two new derivative spaces, the spaces for the zeroth and first order derivatives, such that our sampling space is defined as ⁵:

$$\mathbf{T} = \{\mathbf{T}^{(0)}, \mathbf{T}^{(1)}, \mathbf{T}^{(2)}, \mathbf{T}^{(3)}, \mathbf{T}^{(4)}\}, \quad (100)$$

Similarly, the orientation of our cusp can still be described in terms of the derivatives of ϕ , such that the components of the normal vector, $\mathbf{n} = (n_x, n_y)$ are given by:

$$n_x = 6 \frac{3(2f_{NL}T_2(2f_{NL}TT_{11} - 4f_{NL}T_1^2 + T_{11}) + T_{112}(2f_{NL}T + 1)^2)^2}{(2f_{NL}T + 1)^2(2f_{NL}(TT_{11} - TT_{22} + T_1^2 - T_2^2) + T_{11} - T_{22})} \quad (101)$$

$$+ 2f_{NL}TT_{1111} + 8f_{NL}T_1T_{111} + 6f_{NL}T_{11}^2 + T_{1111}$$

$$n_y = \frac{1}{(2f_{NL}T + 1)^2(2f_{NL}(TT_{11} - TT_{22} + T_1^2 - T_2^2) + T_{11} - T_{22})} \quad (102)$$

$$\times \left[f_{NL} \left(4T(T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122}) \right. \right.$$

$$\left. \left. + T_1^2T_{1112} + 3T_1T_{11}T_{112} + T_2(T_{11}T_{111} + 3T_{11}T_{122} - T_{111}T_{22} - T_{1112}T_2) \right) \right.$$

$$\left. + T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122} + \mathcal{O}(f_{NL}^2) \right].$$

A full expression for n_y can be found in Appendix D.

From here, the process to generate the Local constrained non-Gaussianities follows a straightforward process, as shown in Figure 7. However, the nonlinearity of these constraints means that the average constraint doesn't necessarily need to be part of the constraint manifold, and in fact, in this case, it isn't. As such, in these cases we focus on specific realisations of the field and how different values of f_{NL} affect it, an example of these realisations is shown in

⁵The correlation matrices for the 2D derivative space necessary for this are presented in Appendix B

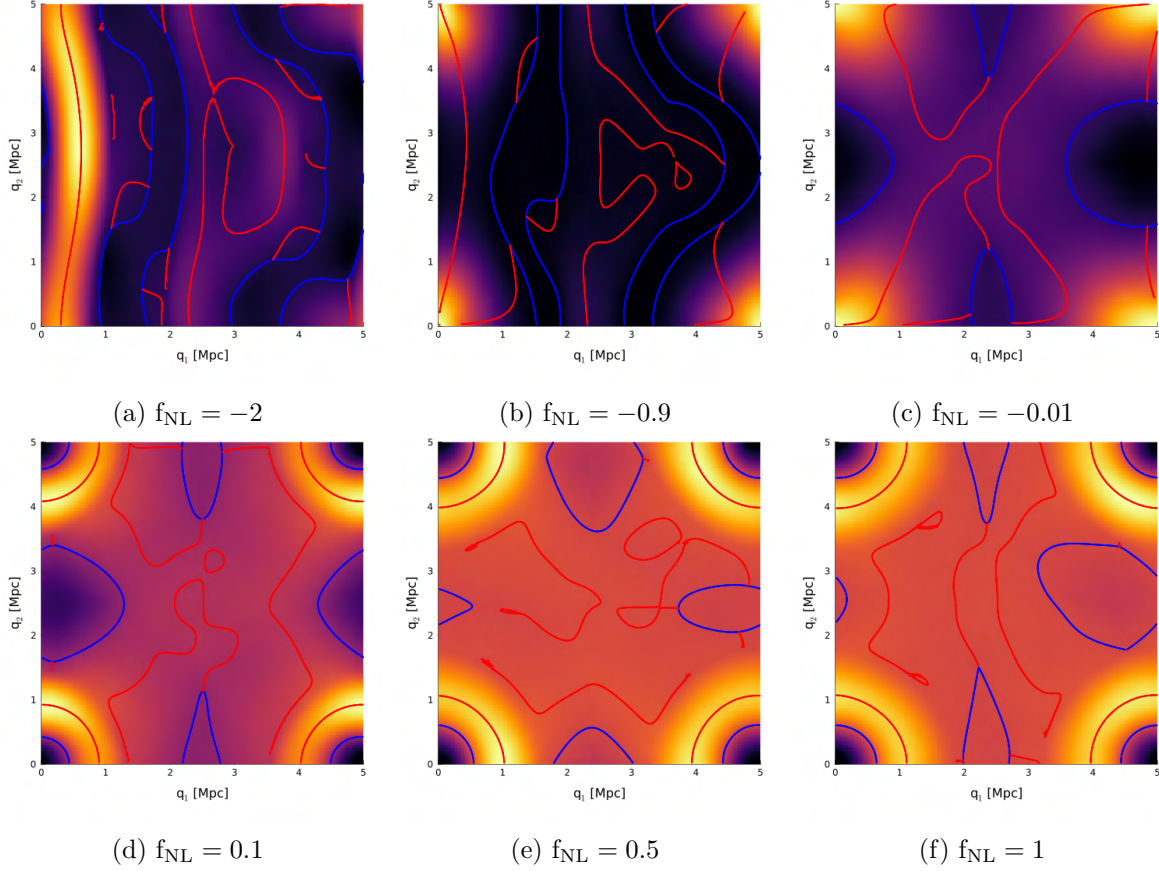


Figure 8. 2D constrained filament for a Local non-Gaussian field, represented in the 1st eigenvalue field. Here we can see how the caustics in Eulerian Space change for a specific realisation of a constrained filament for different values of f_{NL} . Here, the blue lines represent the A_2 caustics, while the A_3 caustics are shown in red. We can see a clearly defined filament at the centre of each field realisation, thus demonstrating the success of our methodology.

Figure 8. In this Figure, we can see how the different values of f_{NL} affect a specific realisation of a 2D Local non-Gaussian field. The range of f_{NL} was chosen to be within the range of possible values expected by the Planck collaboration ($f_{\text{NL}} = -0.9 \pm 5.1$) [7], while showing that this method succeeds to impose a constrained cusp realisation for a wide range of values of f_{NL} . In here, we can see that different f_{NL} values changes the shape of the caustics, which means future studies observing the properties of a large set of these configurations should be able to study and quantify how the different values of f_{NL} affect the statistical properties of a filament. An example of this type of analysis, in this case using stochastic geometry, can be seen in [29].

5.1.2 3D constraints

Having successfully generated the constraints for the 2D case, we can apply the same invariants method to a 3D cosmic wall. Similarly, the deformation tensor is now written as:

$$M = \begin{pmatrix} \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_1} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_2} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_3} \\ \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_2} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_2 \partial x_2} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_2 \partial x_3} \\ \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_3} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_2 \partial x_3} & \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_3 \partial x_3} \end{pmatrix}. \quad (103)$$

From this, we can solve the equations for the 3 invariants of this tensor, obtaining the following constrained conditions:

Local Constrained Wall Conditions:

A₃ conditions ($\lambda_1 = 1/D_+(t)$, $\mathbf{v}_1 \cdot \nabla \lambda_1(\mathbf{q}) = 0$):

$$T_{11} = -\frac{2D_+(t)f_{NL}T_1^2 - 1}{D_+(t)(2f_{NL}T + 1)}, \quad (104)$$

$$T_{111} = \frac{6f_{NL}T_1(2D_+(t)f_{NL}T_1^2 - 1)}{D_+(t)(2f_{NL}T + 1)^2}. \quad (105)$$

Maximum Stretch Condition ($\mathbf{v}_2 \cdot \nabla(\lambda_2 + \lambda_3) = 0$, $\mathbf{v}_3 \cdot \nabla(\lambda_2 + \lambda_3) = 0$):

$$T_{222} = -\frac{1}{(2f_{NL}T + 1)^2} (4f_{NL}^2T^2T_{233} + 12f_{NL}^2TT_2T_{22} + 4f_{NL}^2TT_2T_{33} - 8f_{NL}^2T_2T_3^2 + 4f_{NL}TT_{233} + 6f_{NL}T_2T_{22} + 2f_{NL}T_2T_{33} + T_{233}), \quad (106)$$

$$T_{223} = -\frac{2f_{NL} \left(T_3 \left(-\frac{4f_{NL}T_2^2}{2f_{NL}T+1} + T_{22} + 3T_{33} \right) + TT_{333} \right) + T_{333}}{2f_{NL}T + 1}. \quad (107)$$

Eigenframe Condition ($\frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_2} = \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_2 \partial x_3} = \frac{\partial^2 \Phi_{Local}(\mathbf{x})}{\partial x_1 \partial x_3} = 0$):

$$T_{12} = -\frac{2f_{NL}T_1T_2}{2f_{NL}T + 1}, \quad (108)$$

$$T_{13} = -\frac{2f_{NL}T_1T_3}{2f_{NL}T + 1}, \quad (109)$$

$$T_{23} = -\frac{2f_{NL}T_2T_3}{2f_{NL}T + 1}. \quad (110)$$

First Eigenvalue Condition ($\lambda_1 > \lambda_2 > \lambda_3$):

$$2 f_{NL} (TT_{22} + T_2^2) + T_{22} \leq \frac{1}{D_+(t)}, \quad (111)$$

$$T_{33} < 2f_{NL} (TT_{22} - TT_{33} + T_2^2 - T_3^2) + T_{22}. \quad (112)$$

Similarly, these constraints also fall to the limiting Gaussian case when $f_{NL} = 0$. With these, we have, for the first time, developed a method to study a cosmic wall in a Local non-Gaussian field. While these results are shown only analytically, since due to time constraints, we have not developed our own 3D simulations in this study. The success of the 2D case shown in Figure 8, shows that the application of the method described here to the simulations described in [29] becomes trivial, and we hope to do just that, as well as a full statistical study of the properties of such structures, in future research.

5.2 Defining a Generic Non-Gaussian Random Field

Having successfully extended our constrained methodology to Local type non-Gaussianities, we are now interested in studying how to apply it to a generic non-Gaussian field. To do this, we must first properly define a non-Gaussian Random Field and understand how its properties manifest.

As we mentioned when discussing their properties, as a consequence of Wick's theorem, all the information contained in a Gaussian Random Field can be contained in its power spectrum. However, this theorem no longer applies to the case of non-Gaussian Random Fields (nGRF). To describe the properties of such a field, we need to use higher-order statistics. For this project, we focus on defining the properties of a non-Gaussian field as those contained in its Bispectrum, the Fourier transform of the 3-point correlation function:

$$\begin{aligned} \langle \tilde{f}(\mathbf{k}_1) \tilde{f}(\mathbf{k}_2) \tilde{f}(\mathbf{k}_3) \rangle &= \int d\mathbf{x} d\mathbf{y} d\mathbf{z} \exp(-i(\mathbf{k}_1 \cdot \mathbf{x} + \mathbf{k}_2 \cdot \mathbf{y} + \mathbf{k}_3 \cdot \mathbf{z})) \langle f(\mathbf{x}) f(\mathbf{y}) f(\mathbf{z}) \rangle \\ &= (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3), \end{aligned} \quad (113)$$

using this, we can now describe the non-Gaussian nature of our field. In this project, we consider quadratic non-Gaussian fields. This is the simplest way to generate a field that still preserves the cosmological principle (isotropy and homogeneity) [44]. This can be described by a quadratic kernel, $W[\phi, \phi]$, defined as a double convolution of two GRF realisations with the kernel $K[\mathbf{x}, \mathbf{y}]$, a symmetric function, $K[\mathbf{x}, \mathbf{y}] = K[\mathbf{y}, \mathbf{x}]$ dependent only on $|\mathbf{x}|, |\mathbf{y}|$ and $\mathbf{x} \cdot \mathbf{y}$, which means it can be written as $K[x, y, z]$, with $z = |\mathbf{x} + \mathbf{y}|$:

$$\Phi(\mathbf{x}) = \phi(\mathbf{x}) + f_{NL} \int \int d\mathbf{y} d\mathbf{z} \phi(\mathbf{x} + \mathbf{y}) K[\mathbf{y}, \mathbf{z}] \phi(\mathbf{x} + \mathbf{z}) \quad (114)$$

$$= \phi(\mathbf{x}) + f_{NL} \int \int d\mathbf{y} d\mathbf{z} \phi(\mathbf{y}) K[\mathbf{y} - \mathbf{x}, \mathbf{z} - \mathbf{x}] \phi(\mathbf{z}) \quad (115)$$

$$= \phi(\mathbf{x}) + f_{NL} W[\phi, \phi](\mathbf{x}) \quad (116)$$

Similarly to the Local case, f_{NL} is a free parameter that parameterises the amplitude of the non-Gaussianities, and their bounds can be inferred experimentally from CMB observations [7, 45].

However, we are now interested in seeing how this can be represented in Fourier space, where the statistical properties of this field are easier to study. A generic non-Gaussian Random field Fourier space can be written as:

$$\begin{aligned}\tilde{\Phi}(\mathbf{k}) &= \tilde{\phi}(\mathbf{k}) + f_{NL} \iint d\mathbf{k}_1 d\mathbf{k}_2 \delta(\mathbf{k} - (\mathbf{k}_1 + \mathbf{k}_2)) \tilde{K}[\mathbf{k}_1, \mathbf{k}_2] \tilde{\phi}(\mathbf{k}_1) \tilde{\phi}(\mathbf{k}_2) \\ &= \tilde{\phi}(\mathbf{k}) + f_{NL} \int d\mathbf{k}_1 \tilde{K}[k_1, k_1 - k, k] \tilde{\phi}(\mathbf{k}_1) \tilde{\phi}(\mathbf{k}_1 - \mathbf{k}),\end{aligned}\tag{117}$$

which is just a convolution, where $\tilde{K}$ is the Fourier transform of the original kernel and thus $\tilde{K}[\mathbf{k}_1, \mathbf{k}_2] = \tilde{K}[k_1, k_2, k_3]$, with $k_3 = |\mathbf{k}_1 + \mathbf{k}_2|$ since it will obey the same isotropy and homogeneity conditions.

Plugging in the definition of the non-Gaussian field in equation 117 into the definitions of the Bispectrum in equation 113, we can then see that this can be written as:

$$B_{\Phi}(k_1, k_2, k_3) = 2f_{NL} \tilde{K}[\mathbf{k}_1, \mathbf{k}_2] P(k_1) P(k_2) + Cycle, \tag{118}$$

where cycle refers to the cyclic permutations between k_1, k_2 and k_3 . We can then conclude that similarly to how the power spectrum defines a GRF, the Fourier space kernel will be key to describe a non-Gaussian field.

From its statistical definitions, the bispectrum is only a function of the triangular shape of the 3 points it defines, meaning it is symmetric under k_1, k_2, k_3 permutations (thus the cyclic notation needed here). However, from this definition alone, it is easy to see that $\tilde{K}[\mathbf{k}_1, \mathbf{k}_2]$ does not uniquely define a bispectrum, there can be several different kernels yielding the same result. This is because, from its quadratic definition, the kernel is only symmetric under the exchange of the two vectorial quantities that define it. However, if we specify that it is symmetric under the exchange of its 3 arguments, we can avoid this ambiguity by adding an extra constraint on our kernel, forcing it to be symmetric on k_1, k_2 and k_3 . Adding this extra condition allows us then to further simplify equation 118 as:

$$B_{\Phi}^{Gen}(k_1, k_2, k_3) = 2f_{NL} \tilde{K}[k_1, k_2, k_3] (P(k_1)P(k_2) + P(k_1)P(k_3) + P(k_2)P(k_3)). \tag{119}$$

Further exploiting this freedom of kernel choice we just introduced, we can not only demand that the kernel be symmetric, but also force it to be separable:

$$\tilde{K}[k_1, k_2, k_3] = w_1(k_1)w_2(k_2)w_3(k_3), \tag{120}$$

furthermore, for the purposes of this study, we will be focusing on kernels where $w(k) \propto P(k)^\alpha$, $\alpha \in \mathbb{R}$. From this, we can get a new expression for the quadratic kernel, $W[\phi, \phi]$:

$$W[\phi, \phi](\mathbf{x}) = \int \frac{d\mathbf{k}}{(2\pi)^d} e^{i\mathbf{k} \cdot \mathbf{x}} w_3(k) \int d\mathbf{k}_1 w_1(k_1) w_2(k_1 - k) \tilde{\phi}(\mathbf{k}_1) \tilde{\phi}(\mathbf{k}_1 - \mathbf{k}). \quad (121)$$

We have now defined the general formalism with which we can describe any non-Gaussian Random field we may want to constrain. In order to generate constrained realisations, we are interested in defining our constraints in terms of the initial GRF, ϕ . This way, we can apply the Hoffman-Ribak method on GRF and then use the definition above to generate the constrained non-Gaussian Field from the constrained GRF.

However, extending the constrained Gaussian Random Field formalism is not straightforward. This can be seen in the nature of $W[\phi, \phi]$, which is, in general, quadratic in ϕ , meaning that the constraints generated for it will also need to be quadratic. This breaks the conditions for the Hoffman-Ribak algorithm described in section 3.2. As such, finding a solution that would allow us to use this algorithm for a generic non-Gaussian Random Field proves to be quite challenging.

The Local non-Gaussianities previously studied are an exception to this rule, due to their local nature, corresponding to a kernel defined as $\tilde{K}[k_1, k_2, k_3] = 1$, meaning this quadratic dependency of the field is overcome due to the fact we can separate it into two conditions linear in the field ($\phi^2 = \phi \times \phi$), thus allowing us to use the Hoffman Ribak method again.

However, in general, a kernel is non-Local, meaning the manifestation of the non-Gaussian component of the field in real space can only be understood in terms of the long-range distribution of the field (thus, why most of these are better understood in Fourier space, where this scale dependency becomes obvious). As such, we now aim to find a way to approximate the quadratic behaviour of the constraints in the non-Local non-Gaussian random field ($W[\phi, \phi]$) in terms of operators linear in the field.

5.3 Non-Local Non-Gaussianities

Non-Local non-Gaussianities, as the name indicates, do not manifest themselves locally in the field. As such, we must find new, non-local constraints to describe these fields. However, we must first define these. In general, we will deal with non-Gaussian fields that can be divided into a Gaussian part, a local part and a non-local part:

$$\Phi_{\text{NL}}(\mathbf{x}) = \phi(\mathbf{x}) + f_{\text{NL}} [A\phi^2 + W^{\text{NL}}[\phi, \phi]]. \quad (122)$$

The simplest example of these type of fields are the ones predicted for a single field slow roll inflation model with a standard kinetic term [8, 43, 46], the formalism for these is described in Appendix E, in here we will show how our process can be applied for a generic field and periodically refer to the properties of these fields as an example of an application.

In general, the caustic conditions are still defined in terms of the deformation tensor of the non-local field:

$$M = \begin{pmatrix} \frac{\partial^2 \Phi_{\text{NL}}(\mathbf{x})}{\partial x_1 \partial x_1} & \frac{\partial^2 \Phi_{\text{NL}}(\mathbf{x})}{\partial x_1 \partial x_2} \\ \frac{\partial^2 \Phi_{\text{NL}}(\mathbf{x})}{\partial x_1 \partial x_2} & \frac{\partial^2 \Phi_{\text{NL}}(\mathbf{x})}{\partial x_2 \partial x_2} \end{pmatrix}, \quad (123)$$

where M is the deformation tensor for the two-dimensional case. Meaning we can define the caustic conditions for these fields using the same strategy as before. A full example of this for a purely non-local example can be found in Appendix F. In general, for a field made of a Gaussian part and a non-linear part, defined as a non-local operator on the field $W[\phi, \phi]$ would have the following condition to impose an A_2 caustic:

$$T_{11} = \frac{1 - D_+(t)f_{\text{NL}}W_{11}}{D_+(t)}, \quad (124)$$

where $W_{11} = \frac{\partial^2}{\partial^2 x_1 W[\phi, \phi]}$, similarly to how we have been defining the derivatives of the initial GRF. Thus, using the matrix invariants method, a generic non-Gaussian Random Field can be described by a combination of linear and quadratic constraints on ϕ .

However, this presents a challenge, if we need to describe the field in terms of quadratic constraints based on $W^{\text{NL}}[\phi, \phi]$ and its derivatives, these can no longer be imposed using the Hoffman-Ribak algorithm, since this can only be used for linear constraints of the field.

We are then faced with the choice between two mutually exclusive options: imposing the exact constraints on the field or using the Hoffman-Ribak algorithm. While there exist other, alternative constraining algorithms (for example, directly using an HMC sampler on field realisations), none of them are as computationally effective or produce the theoretical insights of the Hoffman-Ribak algorithm (with other constraining methods, we would lose the ability to, for example, study only the mean field of a realisation). As such, we choose to relax our first requirement and aim to find a way to approximate the quadratic constraints we want to impose in terms of linear operators of the field.

It is important to clarify that these approximations will only be used to obtain the constraints to impose on the initial GRF, ϕ , since the methods used to generate the non-Gaussian realisations of the field based on ϕ don't have the same restrictions as Hoffman-Ribak, meaning we generate this field using the original quadratic expressions. To this end, we will present two possible methods for generating these approximations, how they could be applied in our specific case, and their potential limitations. In this project, we focus on the theoretical implementation of these methods, leaving the practical implementation of these and their analysis for future projects.

5.3.1 Low Rank Approximation

The first method we want to present is the use of Singular Value Decomposition to approximate the quadratic kernel $K[\mathbf{x}, \mathbf{y}]$, shown in equation 114, and which defines the non-Local field, as the sum of the multiplication of two separate kernels:

$$K[\mathbf{x}, \mathbf{y}] \approx \sum_i^n \sigma_i K_i^{(1)}[\mathbf{x}] K_i^{(2)}[\mathbf{y}], \quad (125)$$

this method is known as the Low Rank Approximation.

To understand this method, we must understand how Singular value decomposition works. This is the process through which we can always write an $m \times n$ matrix A as a combination of vectors. In order to see this, we start by studying how the matrix A can be decomposed into 3 separate matrices:

$$A = USV^T, \quad (126)$$

where U is an $m \times m$ orthogonal matrix, V is a $n \times n$ orthogonal matrix and S is $m \times n$ diagonal matrix. We can see this as a more generalised case of diagonalising a square matrix, where in that case the S matrix would simply contain its eigenvalues of the diagonal matrix, and U and V would represent the eigenvector matrices. In general, we call the values in matrix S the singular values of A . Each vector represented in U and V will then be associated with a single singular value, we call these singular vectors.

We can organise the matrix S so that the singular values are in descending order, meaning the first singular value contains the most information about the matrix, with this decreasing as we go down the matrix. We can then write this matrix more explicitly, allowing us to observe the usefulness of this statement. Using this, we can write the matrix in terms of the sum of the components of S :

$$A = \sum_r^{\min(m,n)} s_i \mathbf{u}_i \mathbf{v}_i^T \quad (127)$$

where s_i are the singular values of the matrix and $\mathbf{u}_i$ and $\mathbf{v}_i$ are the singular vectors associated it. In effect, what this decomposition has achieved is to find a way to express a rank 2 tensor in terms of the sum of multiples of rank 1 tensors. Similarly, we can apply the same method for higher rank tensors [47].

The idea behind the Low Rank Approximation is that we can make use of a Singular Value Decomposition and express most of the information contained in a higher rank tensor in terms of lower order ones by keeping only the first k values of its singular value decomposition.

Using this strategy, we would be able to express the non-Local field in terms of the 2 separate convolutions of a kernel with the field:

$$W^{\text{NL}}[\phi, \phi] = \sum_i^k \sigma_i \left(\int d\mathbf{y} K_i^{(1)}[\mathbf{y}] \phi(\mathbf{x} + \mathbf{y}) \right) \left(\int d\mathbf{z} K_i^{(2)}[\mathbf{z}] \phi(\mathbf{x} + \mathbf{z}) \right) \quad (128)$$

where k is the number of principal values we chose to keep. From these, we can write constraints of the field in terms of these convolutions and their derivatives. These constraints are linear and thus can be used with the Hoffman-Ribak algorithm.

As an example, we can study the kernel K_I , used to get one of the components of orthogonal and equilateral non-Gaussian fields (as seen in appendix E):

$$K_I[\mathbf{x}, \mathbf{y}] = \int \int \frac{d\mathbf{k}_1 d\mathbf{k}_2}{(2\pi)^{2d}} e^{i\mathbf{k}_1 \cdot \mathbf{x}} e^{i\mathbf{k}_2 \cdot \mathbf{y}} \tilde{K}_I(k_1, k_2, k_1 + k_2) \quad (129)$$

$$= \int \int \frac{d\mathbf{k}_1 d\mathbf{k}_2}{(2\pi)^{2d}} e^{i\mathbf{k}_1 \cdot \mathbf{x}} e^{i\mathbf{k}_2 \cdot \mathbf{y}} \frac{P(k_1 + k_2)^{2/3}}{(P(k_1)P(k_2))^{1/3}}. \quad (130)$$

We can solve this explicitly for the power spectrum form shown in section 4.3, we can get the explicit form:

$$K_I[\mathbf{x}, \mathbf{y}] = \int \int \frac{d\mathbf{k}_1 d\mathbf{k}_2}{(2\pi)^{2d}} e^{i\mathbf{k}_1 \cdot \mathbf{x}} e^{i\mathbf{k}_2 \cdot \mathbf{y}} \frac{k_1 k_2}{|\mathbf{k}_1 + \mathbf{k}_2|^2} \times \exp \left[-\frac{2}{3} R_s^2 |\mathbf{k}_1 + \mathbf{k}_2|^2 + \frac{1}{3} R_s^2 (k_1^2 + k_2^2) \right]. \quad (131)$$

To apply this method, we want to numerically evaluate this integral for a grid of $\mathbf{x}$ and $\mathbf{y}$ values, which would take the form of a rank 4 tensor for a 2D kernel. Figure 9 shows how this method works in practice for the kernel represented above. By finding the real-space representation of the Fourier kernel, we can approximate it to obtain linear constraints. While these would be different types of constraints from the ones used previously, we will be able to easily implement them into our HMC sampler, using the expression for the correlation function between constraints ξ_{ij} , shown in equation 56 to find the correlation matrices necessary to sample the new constraint spaces.

This method presents a clear path towards effectively generating constraints for non-Local non-Gaussian fields. However, this might present some drawbacks. Firstly, we can consider the computational power required to generate the new kernels. Each kernel will need to be generated using a $N \times N \times N \times N$ grid, where N represents the grid size we choose for our simulations, the computational power necessary to generate these matrices will then be great. However, this will not be a major issue, since these matrices need to be calculated only once for a given grid size, and their symmetries can be used to further simplify the calculations.

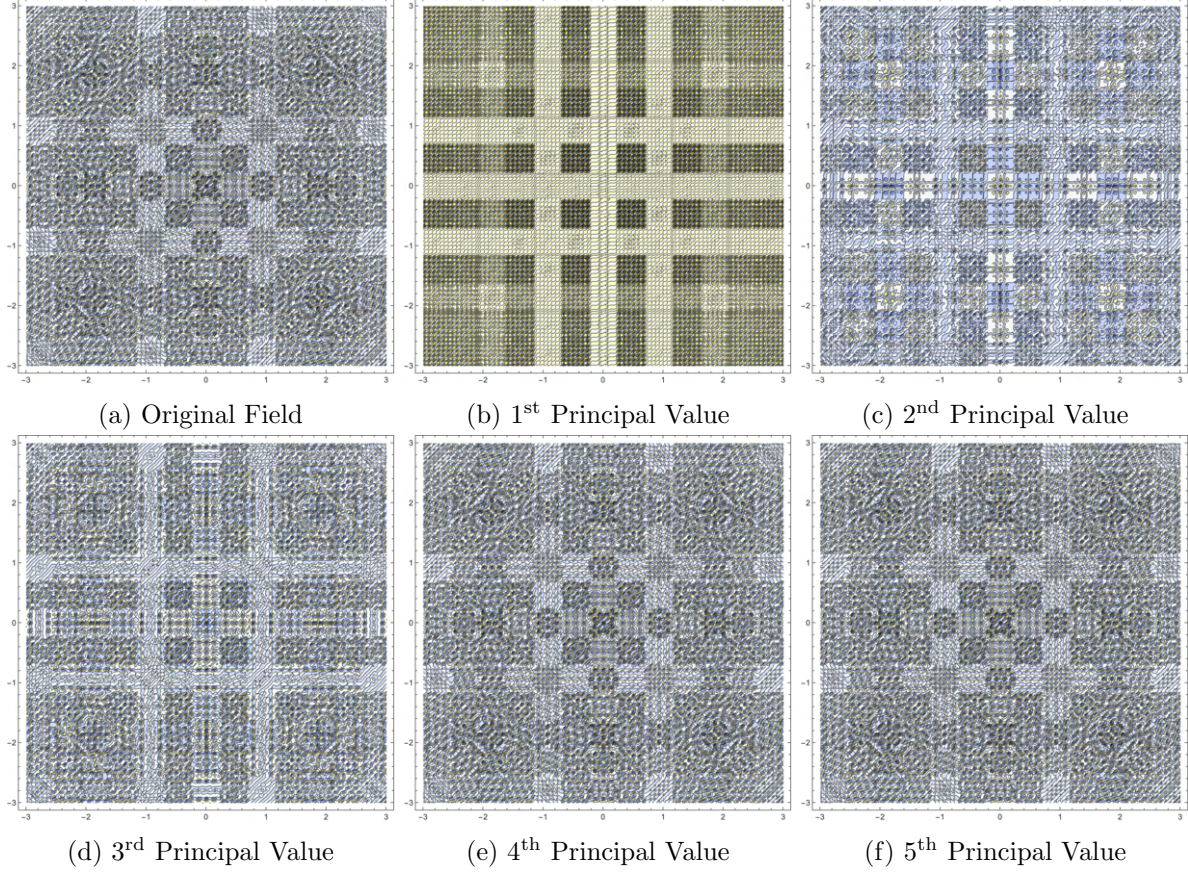


Figure 9. Field reconstruction of a 2D representation of the 2D K_I kernel. In this figure, we have a 2-dimensional representation of the K_I kernel. In here, we reduced the rank 4 tensor into a rank 2 one by representing it in terms of coordinates defined by both components of the $\mathbf{x}$ and $\mathbf{y}$ values. Figure 9a shows the exact form of this kernel, while the following figures show the results of the Low Rank Approximation up to the k^{th} term. We can see that this approximation requires only 4 terms to reproduce the kernel to a high degree of accuracy.

Similarly, with this method, we lose the physical intuition for what our constraints correspond to, in contrast to what we had with the derivative constraints we used before. While this was never a requirement for our constraint conditions, it is a drawback if we need to correct errors in an implementation of this method, or to check whether the implementation was successful, since we have no way to check this apart from confirming the final result and seeing whether or not we have a cusp point at the centre of the field.

As such, it would be interesting to consider possible alternatives that address these concerns and what other drawbacks they might imply. One such alternative is the Taylor Expansion Approximation.

5.3.2 Taylor Expansion of the field

For this method, we first consider how we define our initial GRF. The properties of this field are all contained in the Power Spectrum defined in section 4.3, where $P(k) \propto k^{-3}$, with a

cut-off frequency $R_s = 1$ at high frequencies. This means we can approximate the field as slowly varying over small distances ($R < R_s$). From this, we assume we can approximate this field using a Taylor expansion:

$$\phi(\mathbf{x} + \mathbf{y}) \approx \phi(\mathbf{x}) + \sum_i y_i \partial_i \phi(\mathbf{x}) + \sum_{ij} \frac{1}{2} y_i y_j \partial_i \partial_j \phi(\mathbf{x}). \quad (132)$$

From this we can rewrite the expression for $W[\phi, \phi]$ as:

$$W[\phi, \phi] \approx \phi(\mathbf{x}) \int \int d\mathbf{y} d\mathbf{z} K[\mathbf{y}, \mathbf{z}] \phi(\mathbf{x} + \mathbf{z}) \quad (133)$$

$$+ \sum_i \partial_i \phi(\mathbf{x}) \int \int d\mathbf{y} d\mathbf{z} K[\mathbf{y}, \mathbf{z}] \phi(\mathbf{x} + \mathbf{z}) y_i + \dots \quad (134)$$

This would allow us to write the field in terms of linear operators on ϕ . From this, we would be able to describe the constraints for structures of the cosmic web only in terms of linear operators of ϕ as well:

$$C^{(ijk\dots)}[\phi] = \int \int d\mathbf{y} d\mathbf{z} K[\mathbf{y}, \mathbf{z}] \phi(\mathbf{x} + \mathbf{z}) y_i. \quad (135)$$

While simpler, this approximation retains some of the physical intuition about what the different constraints mean, retaining a connection with an analytical expression of the kernel. Similarly, evaluating this integral should be less computationally expensive, although, as we mentioned before, this is not a major concern. However, this approximation also presents some drawbacks, namely, in this case, determining how accurate it needs to be to produce the necessary constraints is not straightforward. The accuracy of this approximation depends on both the value of f_{NL} and the order of the approximation.

This raises the next step in the analysis of these two alternative approximations: what degree of accuracy do we require in order to produce accurate constraints on the field? Both of these approximations have been created to capture the constraints we impose on our field realisation via the Hoffman-Ribak algorithm's method for constraining the base GRF, ϕ . This is due to the fact that the linearity requirement is only a concern when generating ϕ , once the constraints have been imprinted into this field, we are free to use the quadratic methods to generate the non-Gaussian realisation. As such, future studies must explore this to determine how accurate either of our approximations need to be to produce the desired properties in the field (in our case, an A_3 cusp point at its centre).

6 Conclusion

The caustic skeleton method, together with constrained random field theory, offers a new avenue to study the evolution of the universe and the formation of its structures directly from the conditions of the primordial universe. However, so far this method has only ever been applied to Gaussian Random Fields, excluding from this analysis the possibility of identifying many early universe theories.

The goal of this project has been to extend the existing methodology to allow us to study non-Gaussian fields. Throughout this project, we have succeeded in doing this by:

1. Developing, for the first time, a method to identify the exact constraints for a Local non-Gaussian field in both two and three dimensions, showing how this would work for an A_3 type caustic and demonstrating the success of this with a 2D simulation.
2. Showing how, using the matrix invariants method, it is possible to obtain the constraints for generic non-Local non-Gaussian fields in terms of quadratic constraints of the initial Gaussian Random Field, which will not be compatible with the Hoffman-Ribak algorithm used to impose the constraints until now.
3. Providing two alternative methods of approximating for the non-Local constraints that would allow us to still work with the Hoffman-Ribak method, using either a Taylor expansion of the GRF or a Low Rank Approximation of the real space kernel that describes the non-Gaussian behaviour.

These results, while successfully fulfilling the goal of developing a method to study constrained non-Gaussian fields, open several future research avenues.

The most urgent task for any study would be to implement the approximation methods to generate constrained realisations for different types of non-Gaussian fields described in sections 5.3.1 and 5.3.2. The first step on this would be to develop constrained realisations of the different kernels presented for the simple models for Equilateral and Orthogonal non-Gaussianities presented by [48] (and described in Appendix E).

Once this has been achieved for both methods, the first thing will be to answer the question we left unanswered in section 5.3.2, how accurate will these approximations need to be? While we have proposed two methods that will be able to generate these constraints, it is not clear how accurate either of these approximations will need to be to produce accurate constraints. In order to develop a successful methodology, it would be necessary to determine to what order these must be used in order to produce accurate representations of a structure. Finding what approximation can be used to the lowest order would give us the most effective constraining method since the lower the order, the fewer parameters we need to sample, making it easier for the HMC sampler to accurately sample the manifold.

The last direct continuation of this study would then be to produce a full analysis of how different types of non-Gaussianities affect 2D filaments and 3D walls across different values of

f_{NL} , examining how they affect growth and evolution. This can be done through a statistical analysis, studying properties across several constrained realisations. An interesting approach would be the use of stochastic geometry to study aspects such as the number density of wall centres or the area density of the cosmic walls in a 3D field, allowing us to establish a parallel with the results presented in [29].

Moving towards implementing this methodology into the wider field, the generality of the methods here described would allow them to be integrated in existing pipelines, such as the GENGARS simulation [23]. The formalism used by this simulation to generate accurate initial conditions for N-Body simulations from any arbitrary bispectrum seems to be well-suited to the constrained methodology shown in this report, and as such, it would be interesting to see how this code could be used to improve this simulation and produce more accurate analyses.

Ultimately, once we succeed in implementing this method, it opens the door to exploring a new avenue for studying different early universe theories, as we discussed in our introduction. Once we have succeeded in creating a constraining algorithm and studying the impacts of different types of non-Gaussianities, we can take this one step further, working towards how we can use this method to identify statistical probes that can be compared with observational data, thus being able to test the predictions of early universe theories against what is observed and thus help constrain some of their parameters.

Overall, with this method, we aimed to provide the stepping stones towards the development of new studies to probe the cosmic web from a different perspective. Having succeeded in our goals, we have opened the doors to new and exciting research avenues that will hopefully help us learn more about the universe.

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Appendices

Note — Appendices are provided for completeness only and any content included in them will be disregarded for the purposes of assessment.

A Catastrophes in the cosmic web

Through the shell-crossing conditions we have started to differentiate between different crossings between multistream regions, however we still don't have a method to map these different crossings to different structures. Once again, we are interested into finding ways to study and differentiate the behavior at different crossing points. The caustic skeleton method takes advantage of catastrophe theory to do just that, identifying how different catastrophe families relate to the shell crossing conditions and this can be used to map different structures. While the details of these calculations are not relevant, it is important to get an intuition to how catastrophe theory works in order for us to understand why this mapping is a valid option.

Catastrophe theory is the study and classification of critical points of a function, points defined as [49]:

$$\nabla f(\mathbf{x})|_{x_0} = 0. \quad (136)$$

This is better seen in a 1D functions $f(x)$. In this case, critical points correspond to the places where the first derivative of this function is zero. In this case, we know that we can classify this point as either being a maximum, a minimum or a saddle point of the function, depending on whether the second derivative is positive, negative, or zero. We can see how the definition of a critical point can be expanded to higher dimensions through equation 136 and how the classification of these points will also be dependent on a generalization of the second derivative, its Hessian, $H(\mathbf{x}) = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$.

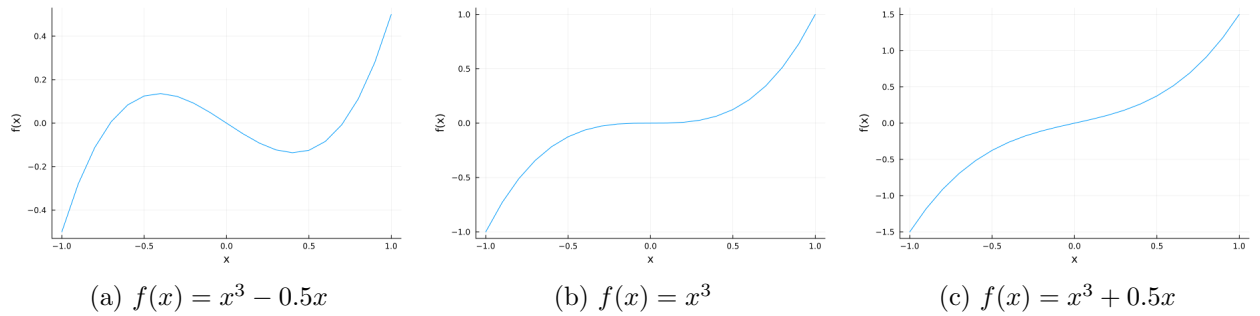


Figure 10. The unwrapping of x^3 , a small perturbation ($a = \pm 0.5$) of this function qualitatively changes the behavior of this function near the origin

If we go back to our 1D example, we can focus on what happens when we have a saddle point. Let's start with the simplest example we can have, a function $f(x) = x^3$, where we will have a saddle point at the origin. If we decided to perturb this function, we can describe this as adding a linear term in x at the origin $f(x) = x^3 - ax$. However, as Figure 10 shows, a change in sign of the constant a will alter the behavior of this function, changing the position of the maximum and minimum close to the origin. This happens because x^3 is not a critically stable function, meaning a small perturbation in it will change its behavior, in other words it represents a singularity at the origin. However, our perturbed function is, since for $f(x) = x^3 - ax$, any small perturbation will keep its behavior close to the origin the same, as long as we know the value of the parameter a . We call this process of turning a critically unstable function stable unfolding. If this unfolding uses the smallest possible number of parameters, we call this universal unfolding and these are unique to each type of singularity.

In general, critically unstable functions will occur whenever $\det(H) = 0$. The focus of catastrophe theory is that for functions that can be approximated as a polynomial (ie. any function can be Taylor expanded), there is only a limited number of universal unfoldings (and thus a limited number of singularities), for a given dimensional space, we call these the elementary catastrophes. There are 5 elementary catastrophes in 2D and 7 in 3D.

This is relevant when looking for our classification method, because our Eulerian field can be written in terms of $\mathbf{x}(q) = \nabla F$ such that:

$$F(\mathbf{q}) = \frac{(\mathbf{q} - \mathbf{x})^2}{2} + D_+(t)\psi(\mathbf{q}), \quad (137)$$

meaning that:

$$\frac{\partial F}{\partial \mathbf{q}_i} = 0 \Leftrightarrow \mathbf{x}(\mathbf{q}) = \mathbf{q} - \nabla \psi = \mathbf{q} - \mathbf{s}_t(\mathbf{q}) \quad (138)$$

and thus allowing us to recover the expression for the shell crossing condition for the eigenvalues of the deformation tensor:

$$\left| \frac{\partial F}{\partial \mathbf{q}_i \partial \mathbf{q}_j} \right| = 0 \Leftrightarrow |I + M| = 0 \quad (139)$$

From this we can see how our shell-crossing conditions along the direction of crossing (equation 10) correspond exactly to the conditions for critical instability for the function $F(\mathbf{q})$. While we are not interested in the unfolding of this function, we can take advantage of this to use the different catastrophe types to identify the different structures in the cosmic web, we call these structures caustics, the physical manifestation of the different catastrophes in the cosmic web.

B Correlation Matrices for a 2D derivative space

In 2D the correlation matrices are given by calculating:

$$\langle (\partial_x^{m_1} \partial_y^{n_1} \phi) (\partial_x^{m_2} \partial_y^{n_2} \phi^*) \rangle \quad (140)$$

$$= \frac{1}{2\pi} (-1)^{m_1+n_1} (i)^{m_1+n_1+m_2+n_2} \sigma_{\frac{1}{2}(m_1+n_1+m_2+n_2)}^2 \int_0^{2\pi} \sin^{n_1+n_2}(\theta) \cos^{m_1+m_2}(\theta) d\theta, \quad (141)$$

where σ_i are the moments of the GRF ϕ . This calculation gives the following derivatives for the full derivative space:

$$\Sigma_{00} = \sigma_0^2 \quad (142)$$

$$\Sigma_{02} = -\frac{\sigma_1}{2} \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} \quad (143)$$

$$\Sigma_{04} = \frac{\sigma_2}{8} \begin{pmatrix} 3 & 0 & 1 & 0 & 3 \end{pmatrix} \quad (144)$$

$$\Sigma_{11} = \sigma_1^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (145)$$

$$\Sigma_{13} = \sigma_2^2 \begin{pmatrix} 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix} \quad (146)$$

$$\Sigma_{22} = \frac{\sigma_2^2}{8} \begin{pmatrix} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix} \quad (147)$$

$$\Sigma_{24} = -\frac{\sigma_3^2}{16} \begin{pmatrix} 5 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 5 \end{pmatrix} \quad (148)$$

$$\Sigma_{33} = \frac{\sigma_3^2}{16} \begin{pmatrix} 5 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 5 \end{pmatrix} \quad (149)$$

$$\Sigma_{44} = \frac{\sigma_4^2}{128} \begin{pmatrix} 35 & 0 & 5 & 0 & 3 \\ 0 & 5 & 0 & 3 & 0 \\ 5 & 0 & 3 & 0 & 5 \\ 0 & 3 & 0 & 5 & 0 \\ 3 & 0 & 5 & 0 & 35 \end{pmatrix} \quad (150)$$

These can be combined into the following correlation matrix:

$$\Sigma = \Sigma^{(0,1,2,3,4)} = \begin{pmatrix} \Sigma_{00} & 0 & \Sigma_{02} & 0 & \Sigma_{04} \\ 0 & \Sigma_{11} & 0 & \Sigma_{13} & 0 \\ \Sigma_{02} & 0 & \Sigma_{22} & 0 & \Sigma_{24} \\ 0 & \Sigma_{13} & 0 & \Sigma_{33} & 0 \\ \Sigma_{04}^T & 0 & \Sigma_{24} & 0 & \Sigma_{44} \end{pmatrix}, \quad (151)$$

C 2D Rotation Matrices

When we derive the constraints for both the Gaussian and the Local type non-Gaussian fields in 2 dimensions, we need to rotate them so that the cusp orientation can be fixed. Our job is simplified by considering that each derivative behaves in its own space, obeying an SO(2) symmetry. This means that when we want to rotate a given derivative, this will be given by the expression:

$$T'_i = R_{ij}^T T_j, \quad (152)$$

where R is is the 2D SO(2) rotation matrix:

$$R = \begin{pmatrix} c & -s \\ s & c \end{pmatrix} \quad (153)$$

where $c = \cos(\theta)$ and $s = \sin(\theta)$.

Now, we know each derivative is linearly independent, so if we want to obtain the same result for a second-order derivative, this can be found by the expression:

$$T'_{ij} = R_{ia}^T R_{ib}^T T_{ab}. \quad (154)$$

We can use this expression to obtain the rotation matrix for the second-order derivative space:

$$R_2 = \begin{pmatrix} c^2 & -2cs & s^2 \\ cs & c^2 - s^2 & -cs \\ s^2 & 2cs & c^2 \end{pmatrix} \quad (155)$$

The same process can now be used to obtain the results for the third and fourth order derivatives:

$$T'_{ijk} = R_{ia}^T R_{ib}^T R_{ic}^T T_{abc}, \quad (156)$$

$$T'_{ijkl} = R_{ia}^T R_{ib}^T R_{ic}^T R_{id}^T T_{abcd}. \quad (157)$$

Following the same process as before, we can then use this to find the corresponding rotation matrices:

$$R_3 = \begin{pmatrix} c^3 & -3c^2s & 3cs^2 & -s^3 \\ c^2s & c^3 - 2cs^2 & -2c^2s + s^3 & cs^2 \\ cs^2 & 2c^2s - s^3 & c^3 - 2cs^2 & -c^2s \\ s^3 & 3cs^2 & 3c^2s & c^3 \end{pmatrix} \quad (158)$$

$$R_4 = \begin{pmatrix} c^4 & -4c^3s & 6c^2s^2 & -4cs^3 & s^4 \\ c^3s & c^4 - 3c^2s^2 & -3c^3s + 3cs^3 & 3c^2s^2 - s^4 & -cs^3 \\ c^2s^2 & 2cs(c^2 - s^2) & c^4 - 4c^2s^2 + s^4 & -2c^3s + 2cs^3 & c^2s^2 \\ cs^3 & 3c^2s^2 - s^4 & 3cs(c^2 - s^2) & c^4 - 3c^2s^2 & -c^3s \\ s^4 & 4cs^3 & 6c^2s^2 & 4c^3s & c^4 \end{pmatrix} \quad (159)$$

D Local Non-Gaussianities full expressions

Here you can find the full expression for some of the expressions for the Local-Type non-Gaussianities, too long to be expressed in the main body of the report:

2D Orientation:

$$\begin{aligned}
n_y = & \frac{1}{(2f_{\text{NL}}T + 1)^2 (2f_{\text{NL}} (TT_{11} - TT_{22} + T_1^2 - T_2^2) + T_{11} - T_{22})} \\
& \times \left[16f_{\text{NL}}^4 \left(T^4 (T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122}) \right. \right. \\
& \quad + T^3 (T_1^2T_{1112} + 3T_1T_{11}T_{112} + T_2(T_{11}T_{111} + 3T_{11}T_{122} - T_{111}T_{22} - T_{1112}T_2)) \\
& \quad + T^2 (3T_1^3T_{112} + T_1^2T_2(T_{111} - 6T_{122}) - 3T_1T_2(T_{11}^2 - 2T_{11}T_{22} + 3T_{112}T_2) - T_{111}T_2^3) \\
& \quad \left. \left. - 3TT_1T_2(T_1^2(T_{11} + 2T_{22}) + T_{11}T_2^2) + 12T_1^3T_2^3 \right) \right. \\
& + 8f_{\text{NL}}^3 \left(4T^3 (T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122}) \right. \\
& \quad + 3T^2 (T_1^2T_{1112} + 3T_1T_{11}T_{112} + T_2(T_{11}T_{111} + 3T_{11}T_{122} - T_{111}T_{22} - T_{1112}T_2)) \\
& \quad + 2T (3T_1^3T_{112} + T_1^2T_2(T_{111} - 6T_{122}) - 3T_1T_2(T_{11}^2 - 2T_{11}T_{22} + 3T_{112}T_2) - T_{111}T_2^3) \\
& \quad \left. \left. - 3T_1T_2(T_1^2(T_{11} + 2T_{22}) + T_{11}T_2^2) \right) \right. \\
& + 4f_{\text{NL}}^2 \left(6T^2 (T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122}) \right. \\
& \quad + 3T (T_1^2T_{1112} + 3T_1T_{11}T_{112} + T_2(T_{11}T_{111} + 3T_{11}T_{122} - T_{111}T_{22} - T_{1112}T_2)) \\
& \quad \left. \left. + 3T_1^3T_{112} + T_1^2T_2(T_{111} - 6T_{122}) - 3T_1T_2(T_{11}^2 - 2T_{11}T_{22} + 3T_{112}T_2) - T_{111}T_2^3 \right) \right. \\
& + 2f_{\text{NL}} \left(4T (T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122}) \right. \\
& \quad + T_1^2T_{1112} + 3T_1T_{11}T_{112} + T_2(T_{11}T_{111} + 3T_{11}T_{122} - T_{111}T_{22} - T_{1112}T_2) \left. \right) \\
& \left. + T_{11}T_{1112} - T_{1112}T_{22} + 3T_{112}T_{122} \right].
\end{aligned}$$

E Orthogonal and Equilateral Non-Gaussianities

In this section, we will focus on the simplest cases of non-Local non-Gaussianities: equilateral and orthogonal non-Gaussianities as predicted for a single field slow roll inflation model with a standard kinetic term [8, 43, 46]. The values of f_{NL} for these non-Gaussianities have been measure by the Planck Collaboration and are given as $f_{NL}^{eq} = -26 \pm 47$ and $f_{NL}^{ortho} = -38 \pm 24$.

Equilateral and Orthogonal non-Gaussianities can be defined by their bispectra [50]:

$$\begin{aligned} \frac{1}{6f_{NL}} B_{\Phi}^{Eq} = & -(P(k_1)P(k_2) + Cycle) - 2(P(k_1)P(k_2)P(k_3))^{\frac{2}{3}} \\ & + (P(k_1)^{\frac{1}{3}}P(k_2)^{\frac{2}{3}}P(k_3) + Cycle), \end{aligned} \quad (160)$$

$$\begin{aligned} \frac{1}{6f_{NL}} B_{\Phi}^{Ort} = & -3(P(k_1)P(k_2) + Cycle) - 8(P(k_1)P(k_2)P(k_3))^{\frac{2}{3}} \\ & + 3(P(k_1)^{\frac{1}{3}}P(k_2)^{\frac{2}{3}}P(k_3) + Cycle). \end{aligned} \quad (161)$$

We now want to use these expressions to find the expressions for the fourier space kernel in the form described in equation 120. For this we will use the method described in [48]. As such we will just cite the results from these unless the derivations are directly relevant to the implementation of our constraining method.

To generate the kernel, we start by observing that both our fields can be separated into 3 separated into 3 terms:

- **Local Term** : $B_{Local} = (P(k_1)P(k_2) + Cycle)$
- **Term 1** : $B_I = (P(k_1)P(k_2)P(k_3))^{\frac{2}{3}}$
- **Term 2** : $B_{II} = (P(k_1)^{\frac{1}{3}}P(k_2)^{\frac{2}{3}}P(k_3) + Cycle)$

The local term, we can treat it as we did in the previous section and not worry about it. For terms 1 and 2 we now need to find the kernels that obey the relations:

$$2f_{NL}\tilde{K}(k_1, k_2, k_3)P(k_1)P(k_2) + Cycle = (P(k_1)P(k_2)P(k_3))^{\frac{2}{3}}. \quad (162)$$

$$2f_{NL}\tilde{K}(k_1, k_2, k_3)P(k_1)P(k_2) + Cycle = (P(k_1)^{\frac{1}{3}}P(k_2)^{\frac{2}{3}}P(k_3) + Cycle). \quad (163)$$

We can see that such a term must have the form of :

$$(P(k_1)^{-\alpha}P(k_2)^{-\beta} + P(k_2)^{-\alpha}P(k_1)^{-\beta})P(k_3)^{-(2-\alpha-\beta)}, \quad (164)$$

For term 1, we can see this is straight forward with:

$$\tilde{K}_I(k_1, k_2, k_3) = \frac{P(k_3)^{2/3}}{(P(k_1)P(k_2))^{1/3}}. \quad (165)$$

However for term 2, there are several possible solutions:

$$\tilde{K}_{II}^A(k_1, k_2, k_3) = P(k_3)^{1/3}(P(k_1)^{-1/3} + P(k_2)^{-1/3}). \quad (166)$$

$$\tilde{K}_{II}^B(k_1, k_2, k_3) = P(k_3)(P(k_1)^{-2/3}P(k_2)^{-1/3} + P(k_2)^{-2/3}P(k_1)^{-1/3}). \quad (167)$$

$$\tilde{K}_{II}^C(k_1, k_2, k_3) = P(k_3)^{1/3}(P(k_1)^{-1/3} + P(k_2)^{-1/3}). \quad (168)$$

Giving an overall kernel:

$$\tilde{K}_{II} = (1 - s - t)\tilde{K}_{II}^A + s\tilde{K}_{II}^B + t\tilde{K}_{II}^C, \quad (169)$$

where s and t are the coefficients that determine how these 3 solutions mix. As we mentioned above, from [48] we know that $s = 0$ and $t = \frac{1}{3}$ for equilateral fields and $t = \frac{4}{9}$ for orthogonal fields.

While the knowledge of the how the fourier space kernels is useful, it is interesting to first learn how we can use this information to generate real space representations of these non-Gaussian fields, based on an initial GRF ϕ . Assuming a power spectrum proportional to k^{-3} , we can simplify the equation for the field $\Phi(x)$ in terms of the following differential operators defined in fourier space:

$$\begin{aligned} \partial\phi(\mathbf{x}) &= \int e^{-i\mathbf{k}\cdot\mathbf{x}} k\phi(\mathbf{k})d^3k, & \nabla^{-2}\phi &= - \int e^{-i\mathbf{k}\cdot\mathbf{x}} k^{-2}\phi(\mathbf{k})d^3k, \\ \partial^{-1}\phi(\mathbf{x}) &= \int e^{-i\mathbf{k}\cdot\mathbf{x}} k^{-1}\phi(\mathbf{k})d^3k. \end{aligned}$$

These operators are extremely useful, since they can be computationally calculated in a very efficient manner using fast Fourier transform methods. Using this methods, we can write the expression for the field in Fourier space (equation 117) in terms of these operators and then bring it to real space, returning the useful expressions for our fields:

$$\Phi_{Eq} = \phi + f_{NL} \left[-3\phi^2 + 4\partial^{-1}(\phi\partial\phi) + 2\nabla^{-2}(\phi\nabla^2\phi) + 2\nabla^{-2}(\partial\phi)^2 \right]. \quad (170)$$

$$\Phi_{Ort} = \phi + f_{NL} \left[-9 \phi^2 + 10 \partial^{-1}(\phi \partial \phi) + 8 \nabla^{-2}(\phi \nabla^2 \phi) + 8 \nabla^{-2}(\partial \phi)^2 \right] . \quad (171)$$

Similarly, collecting all we described in this section, the non-Local part of each of these fields can be described in terms of a Fourier space kernel given by:

$$\begin{aligned} \tilde{K}_{eq}^{NL}(k_1, k_2, k_3) &= -2 \tilde{K}_I + 6 \left(\frac{2}{3} \tilde{K}_{II}^A + \frac{1}{3} \tilde{K}_{II}^C \right) \\ &= -2 \tilde{K}_I + 4 \tilde{K}_{II}^A + 2 \tilde{K}_{II}^C \end{aligned} \quad (172)$$

$$\tilde{K}_{eq}^{NL}(k_1, k_2, k_3) = -8 \tilde{K}_I + 10 \tilde{K}_{II}^A + 8 \tilde{K}_{II}^C \quad (173)$$

F 2D Filament Conditions for a Generic Non-Local Field

For a generic field, defined by a purely non-linear kernel:

$$\Phi_{\text{NL}}(\mathbf{x}) = \phi(\mathbf{x}) + f_{\text{NL}} W^{\text{NL}}[\phi, \phi] \quad (174)$$

we can defined the constrained conditions for a filament using the same field invariants based method we have been using before. This returns the following conditions:

Non-Local Constrained Filament Conditions:

A₃ conditions ($\lambda_1 = 1/D_+(t)$, $\mathbf{v}_1 \cdot \nabla \lambda_1(\mathbf{q}) = 0$):

$$T_{11} = \frac{1 - D_+(t) f_{\text{NL}} W_{11}}{D_+(t)}, \quad (175)$$

$$T_{111} = -f_{\text{NL}} W_{111} \quad (176)$$

Maximum Stretch Condition ($\mathbf{v}_2 \cdot \nabla \lambda_2 = 0$):

$$T_{222} = -f_{\text{NL}} W_{222} \quad (177)$$

Eigenframe Condition ($\frac{\partial^2 \Phi_{\text{Local}}(\mathbf{x})}{\partial x_1 \partial x_2}$):

$$T_{12} = -f_{\text{NL}} W_{12} \quad (178)$$

First Eigenvalue Condition ($\lambda_1 > \lambda_2$):

$$f_{\text{NL}} W_{22} + T_{22} \leq \frac{1}{D_+(t)} \quad (179)$$

Where we used the notation $W_{ijk\dots}$ to denote the derivatives of the non-local part of the field, similarly to how we used T's before.

Applying the constrained method for this conditions would seem simple, we would just have to impose new constraints $W_{ijk\dots}$ on the field. What this notation hides is the fact that $W_{ijk\dots}$ would be quadratic constraints, which would break the gaussianity assumption on which the application of the Hoffman-Ribak algorithm depends. Our challenge becomes then whether we can find a suitable approximation for $W_{ijk\dots}$ in terms of a linear operators of the field in a way that would still allow us to impose constraints on the field using the Hoffman-Ribak algorithm.